

**A GEOSPATIAL TECHNIQUE BASED
CYCLONE RISK ASSESSMENT FOR
GREATER CHENNAI**

A PROJECT REPORT

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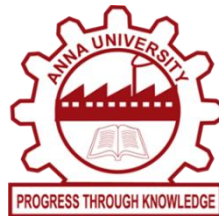
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ABSTRACT

Tropical Cyclones are one of the most powerful and devastating natural disaster that causes severe damage to life and property. In India, most of the Tropical Cyclones develop in the Bay of Bengal. Hence the Eastern Coast of India is more prone to the Cyclones. The Chennai City which is located on the East Coast of India is often hit by the cyclones formed in the Bay of Bengal. Therefore, the study focuses to assess the Cyclone Risk zones using Remote Sensing and GIS techniques. Twelve spatial criteria were selected under the Hazard, Vulnerability and Exposure as the components of risk such as cyclone wind speed, frequency of cyclones, rainfall intensity, slope, elevation, land use land cover, cyclone track range, coastline range, stream range, population density, household density and vegetation cover. The Weighted Overlay method is used to integrate the risk components such as Hazard, Vulnerability and Exposure. The resultant Risk map was divided into three divisions namely high, medium and low. The findings shows that 27% of the total area comes under highly risk zone, 70% under moderate risk zone and 3% under lower risk zone. This information would be useful for the local authorities in advance planning to a disaster which helps to reduce the loss of lives.

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LIST OF ABBREVIATIONS

AHP	Analytical Hierarchy Process
ARCGIS	Aeronautical Reconnaissance Coverage Geographic Information System
DAV	Data Access Viewer
DEM	Digital Elevation Model
DRM	Disaster Risk Management
ESA	European Space Agency
ESRI	Environmental System Research Institute
ENVI	Environment for Visualizing Imagery
FAHP	Fuzzy Analytical Hierarchy Process
FCC	False Color Composite
GIS	Geographic Information System
GPS	Global Positioning System
IBTrACS	International Best Track Archive for Climate Stewardship
IMD	Indian Meteorological Department
LIDAR	Light Detection and Ranging
LULC	Land Use and Land Cover
NASA	National Aeronautical and Space Agency
NCRMP	National Cyclone Risk Mitigation Project
NOAA	National Oceanic and Atmospheric Administration
OSM	Open Street Map

PBF	Protocol Binary Format
SRTM	Shuttle Radar Topography Mission
SWIR	Short Wave Infrared
USGS	United Nations Geological Survey
UTs	Union Territories
VNIR	Visible and Near Infrared
WMO	World Meteorological Organization
WSM	Weighted Sum Model

CHAPTER 1

INTRODUCTION

1.1 GENERAL

A Cyclone is a low-pressure area in the atmosphere where the wind-spiral upward. The word Cyclone has been derived from the Greek word ‘cyclos’ which means ‘coiling of a snake’. It was coined by Henry Paddington, President of the marine courts, and Kolkata during British rule in 1848. Generally, Cyclones are accompanied by strong winds, heavy rain, severe thunder and lightning with storm surges. For the development of cyclones, there needs to be relatively high humidity in the lower and middle troposphere region. This humidity reduces the amount of evaporation in clouds and maximizes the latent heat released because there is more precipitation. The sea surface temperature should be more than 26°C, which acts as a major source of energy. Cyclone will weaken rapidly when they travel over land or colder ocean waters which is termed as landfall.

In the Northern Hemisphere, cyclones rotate in anti-clockwise direction and in the Southern Hemisphere it rotates in clockwise direction. The change in the direction of rotation is due to the Coriolis effect. Cyclones are characterised as Tornadoes, Hurricanes and Typhoons. They are referred to as 'Hurricanes' over the Atlantic Ocean, 'Typhoons' over the Pacific Ocean, and simply as 'Cyclones' over the North Indian Ocean.

Cyclones can live for a long period of time from two to three weeks. Once a disturbance has become a depression, the amount of time it takes to achieve the next stage, that is storm, can take half a day to a couple of days. Atmospheric and oceanic conditions play the major role in determining these events.

The magnitude or intensity of cyclone is measured by Saffir-Simpson Hurricane scale which classifies cyclone on the basis of wind speed between Category 1 to Category 5. On this scale, category 1 is least destructive while category 5 is most destructive cyclone. However, in India a different classification is used to categorise the cyclone.

1.2 CLASSIFICATION OF CYCLONES

The criteria followed by the Indian Meteorological Department (IMD) to classify the low-pressure systems in the Bay of Bengal and in the Arabian Sea as adopted by the World Meteorological Organization (WMO) are given in the TABLE 1.1.

TABLE 1.1 CLASSIFICATION OF CYCLONES

TYPE OF DISTURBANCE	WIND SPEED
Low Pressure Area	< 31 km/h
Depression	31 - 49 km/h
Deep Depression	50 - 61 km/h
Cyclonic Storm	62 - 88 km/h
Severe Cyclonic Storm	89 - 117 km/h
Very Severe Cyclonic Storm	118 - 167 km/h
Extremely Severe Cyclonic Storm	168 - 221 km/h
Super Cyclonic Storm	>222 km/h

1.3 TROPICAL CYCLONES

The Indian subcontinent is one of the worst affected regions of tropical cyclones in the world. It is exposed to nearly 10 percent of the world's tropical cyclones. Of these, the majority of them have their initial genesis over the Bay of Bengal and strike the East coast of India. On an average, five to six tropical cyclones form every year, of which two or three could be severe. More cyclones occur in the Bay of Bengal than the Arabian Sea and the ratio is approximately 4:1. Cyclones occur frequently on both the coasts (the West coast - Arabian Sea; and the East coast - Bay of Bengal). An analysis of the frequency of cyclones on the East and West coasts of India between 1891 and 1990 shows that nearly 262 cyclones occurred (92 of these severe) in a 50 km wide strip above the East coast. Less severe cyclonic activity has been noticed on the West coast, where 33 cyclones occurred the same period, out of which 19 of were severe.

Cyclones occur in the months of May-June and October-November. Cyclones of severe intensity and frequency in the North Indian Ocean are bi-modal in character, with their primary peak in November and secondary peak in May. The disaster potential is particularly high during landfall in the North Indian Ocean due to the accompanying destructive wind, storm surges and torrential rainfall. Of these, storm surges cause the most damage as sea water inundates low lying areas of coastal regions and causes heavy floods, erodes beaches and embankments, destroys vegetation and reduces soil fertility.

1.4 HISTORICAL CYCLONE EVENTS

Over the past decade, the Chennai city has experienced several significant cyclonic events, some of which have caused widespread devastation. Here's an overview of some notable cyclones that have impacted Chennai in the past decade. These cyclonic events underscore the vulnerability of coastal cities like Chennai to the impacts of climate change induced extreme weather events. The historical cyclone events are mentioned in TABLE 1.2.

TABLE 1.2 HISTORICAL CYCLONE EVENTS

YEAR	CYCLONE NAME	WIND SPEED (km/h)	CYCLONE TYPE
2011	Thane	145	Very severe cyclonic storm
2012	Nilam	85	Cyclonic storm
2016	Vardah	130	Very severe cyclonic storm
2020	Nivar	120	Very severe cyclonic
2022	Mandous	95	Severe cyclonic storm
2023	Michaung	100	Severe cyclonic storm

1.5 National Cyclone Risk Mitigation Project (NCRMP),2010

The government of India has introduced the National Cyclone Risk Mitigation Project (NCRMP) to address the issues of cyclone risk in the country. It aims at taking suitable structural and non-structural measures to mitigate the effects of cyclones in the coastal states and UTs of India.

Based on the frequency of occurrence of cyclones, population and the existing institutional mechanism for disaster management, cyclone vulnerable states are divided into two categories.

Category I: Higher vulnerability States- Andhra Pradesh, Gujarat, Odisha, Tamil Nadu and West Bengal.

Category II: Lower vulnerability States- Maharashtra, Goa, Karnataka, Kerala, Daman & Diu, Puducherry, Lakshadweep and Andaman & Nicobar Islands.

Phase I began with Andhra Pradesh and Odisha in 2011. Other States/UTs will be covered progressively.

The Objectives of NCRMP,

- i) To improve Early Warning Dissemination Systems
- ii) Enhanced capacity of local communities to respond to disasters
- iii) To improve access to emergency shelter, evacuation and protection against wind storms, flooding and storm surge in highly risk areas.
- iv) To strengthen Disaster Risk Management (DRM) capacity at central, state and local levels in order to enable mainstreaming of risk mitigation measures into the overall development agenda.

1.6 AIM AND OBJECTIVE

The study aims to develop a comprehensive Cyclone Risk map using Geospatial Techniques to quantify the degree of risk of Cyclone for Chennai in Tamil Nadu.

1. To collect the geospatial datasets such as cyclone wind speed, frequency of occurrences, rainfall intensity to develop a hazard zone map.
2. To identify the elements at risk and create a map of vulnerability and exposure.
3. To integrate the parameters of hazard, vulnerability and exposure to produce the risk map for the study area.

1.7 NEED FOR THE STUDY

- To identify local and regional areas that are at high threat to cyclones.
- To alert the local community and to provide precise data to decision makers.
- To understand the likelihood and assess the risk which helps in planning for response and recovery.
- To create evidence-based evacuation plans and mitigation strategies.

1.8 CONCLUSION

From this chapter, an overview of the Cyclones, its classification, the Historical Events and the Objective of the study are explained.

CHAPTER 2

REVIEW OF LITERATURE

2.1 GENERAL

This chapter discusses the literature that have been reviewed in order to conduct the entire study.

2.2 REVIEW OF LITERATURE

Debabrata Ghorai et al., (2017) The study focuses on the Mapping of Cyclone Vulnerability zones along the Tamil Nadu coast using Remote Sensing and GIS techniques with the four physical parameters such as elevation, wind speed, historical cyclone event, and land use/land cover. They performed Weighted Sum Model (WSM) by integrating all the parameters to achieve the cyclone vulnerability zone map. The result revealed that very high (35.15%), high (24.23%), moderate (31.57%), low (7.11%), and very low (1.93%). The study summarized that the northern coastal taluks are more vulnerable than central and southern coastal taluks of Tamil Nadu coast.

Jacinth Jennifer et al., (2018) The present study developed the vulnerability assessment of Cuddalore district in Tamil Nadu from potential cyclones. Mapping of cyclone vulnerability zone has been carried out with six physical parameters such as elevation, wind speed, historical cyclone event, significant wave height, precipitation, and LULC. They used regular rank and weight method to integrate the above parameters to achieve the cyclone vulnerability zone map. They concluded that the coastal areas of Cuddalore district are more vulnerable than the interior areas.

Stuart Phinn et al., (2018) The study developed and evaluated a multi-criteria approach for mapping the risk levels of areas to tropical cyclone in Sarankhola Upazila, in Coastal Bangladesh. A spatial layer was created for each criterion namely Vulnerability and Exposure, Hazard, Mitigation Capacity and weighting was conducted through the Analytical Hierarchy Process (AHP). The results indicate that eastern, south and south western parts of the study area are at high risk to cyclones.

Ajim Ali S K et al., (2019) A Risk map was developed integrating the layers of vulnerability, hazard and mitigation capacity. They applied a multi-criterion decision-making technique to assess the block-level risk to cyclone in the Sundarban region, India. The weight-based analysis reveals that nearness to coastline and distance from cyclone tract have the greatest vulnerability and exposure, windspeed has the highest hazard score, and on the other hand, nearness to cyclone shelter and cyclone awareness, has utmost mitigation capacity to cyclone risk analysis. The end result shows that, out of 19 blocks in the Sundarban, nearly half of the blocks are with very high to moderate cyclone risk.

Muhammad Al-Amin Hoque et al., (2019) This study aims to produce a risk map for Eastern Coastal Region of Bangladesh by identifying the areas under high cyclone risks to support mitigation strategies. In total there were 14 spatial criteria under three risk components namely vulnerability and exposure, hazard and mitigation. The weighting was conducted following the AHP. The results demonstrates that 9% as very-high risk zone and 27% as high-risk zone, 25% as moderate risk zone, 39% as low and very low risk zones.

Akhtar Alam et al., (2019) The aim of the study is to examine the spatial patterns of cyclone risk in the Cox's Bazar district and Rohingya refugee camps of Bangladesh. Here in this case study, they use parameters representing the hazard, exposure, and vulnerability as the components of risk. The selected parameters were analyzed and integrated through AHP and GIS. The results exhibited that the northwestern and southern peripheral areas exhibited higher risk than central and northern parts of Cox's Bazar.

Bayes Ahmed et al., (2021) The study developed a comprehensive tropical cyclone vulnerability mapping by applying Fuzzy Analytical Hierarchy Process (FAHP) and integrating the physical, social and mitigation capacity categories as the components of vulnerability. The results indicate that the southern and south-eastern peripheral areas exhibit higher vulnerability to tropical cyclones since these areas comprise low elevation, gentle slope, closeness to the sea, a high number of historical cyclone tracks.

Mostafizur Rahman et al., (2023) The study highlights the regions of exposure to cyclones in Taltali, a southern coastal Upazila of Bangladesh. To quantify the exposure, they used 6 criteria including elevation, slope, proximity to major water bodies, population density, proximity to cyclone tracks and land use land cover and produced maps focusing on each criterion. All the analysis was done in a GIS environment using the AHP method. The overall exposure map revealed that about 60 % of the study area was quantified as very highly exposed to cyclones.

Chowdhary A et al., (2023) The research aimed to assess the vulnerability of cyclones in Bangladesh by employing GIS technology, remote sensing, and the AHP method. The primary objective of this study was to identify susceptible areas through data collection on historical cyclone occurrences, DEM, Slope, population density, cyclone shelter locations. The results revealed that approximately 46% of the total area comes under very highly risk zone, 24% moderate risk zone, 30% low risk zone.

Subbarayan Saravanan et al., (2023) The present study exhibits the cyclone risk map for the Tamil Nadu Coast by utilizing four distinct components: hazards, exposure, vulnerability, and mitigation. The entire coast was divided into five zones. They employed AHP technique to compute an integrated risk index considering 16 spatial variables. The resultant risk map demonstrates that zone 1 and 2 are high risk zones.

2.3 CONCLUSION

From the review of literature, it is evident that Satellite Remote Sensing and GIS technology plays a major role in assessing the risk components of a cyclone.

CHAPTER 3

STUDY AREA AND ITS DESCRIPTION

3.1 GENERAL

This chapter deals with the study area and its description.

3.2 LOCATION OF THE STUDY AREA

The area chosen for the study is Greater Chennai which lies between 12°51' N to 13°15' N and 80°8' E to 80°20' E. It is the largest Municipal Corporation in Tamil Nadu with a total area of 426 sq.km. It is located on the south eastern coast of India in the north eastern part of Tamil Nadu on the Eastern Coastal Plain. The study area is shown in FIGURE 3.1.

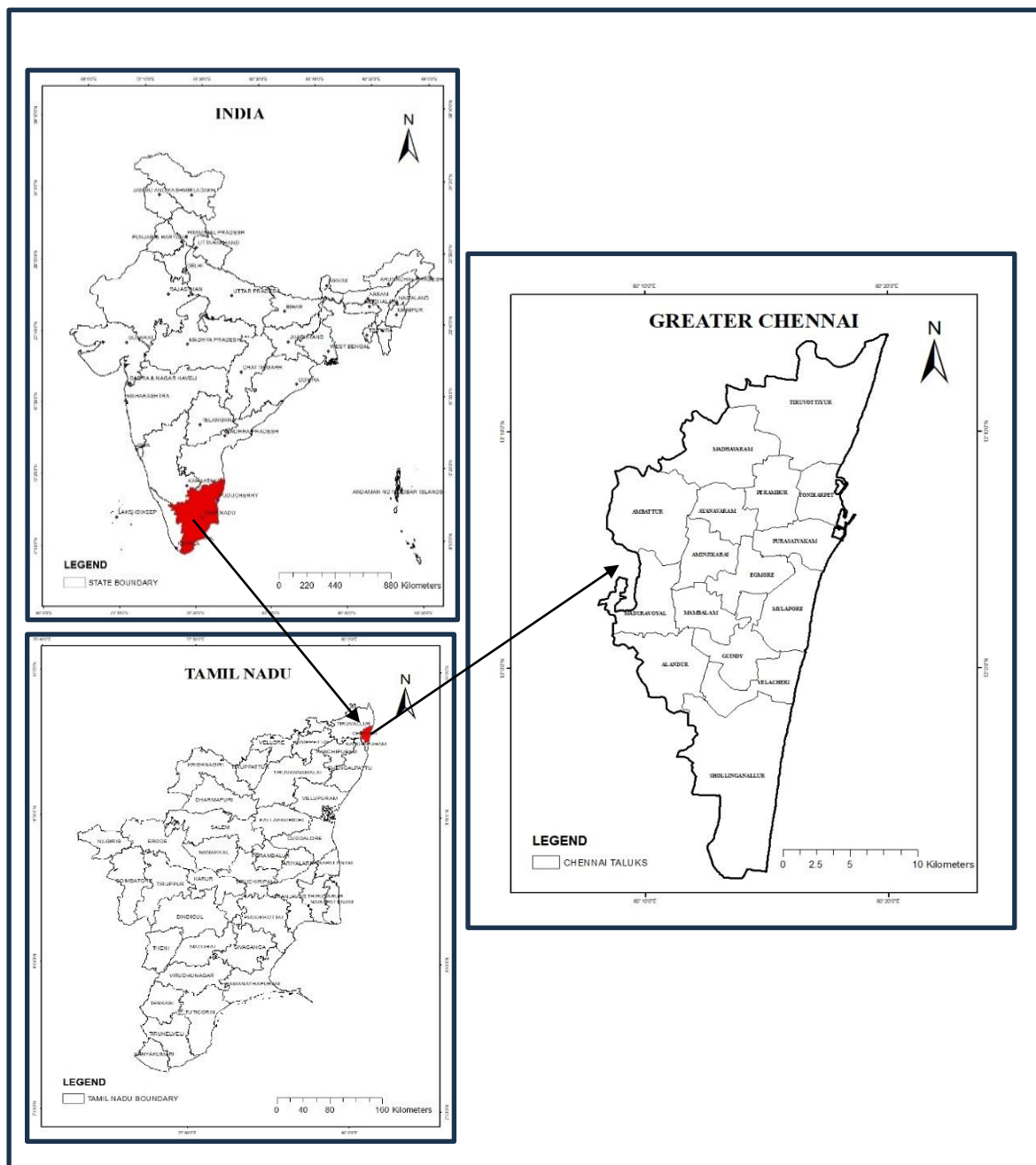


FIGURE 3.1 STUDY AREA

3.3 DESCRIPTION OF THE STUDY AREA

The Greater Chennai Corporation is a local government for the Capital City in Tamil Nadu. In October 2011, 42 small local bodies including 9 municipalities, 8 town panchayats and 25 village panchayats were merged with Chennai Corporation taking up the area from 176 sq. km to 426 sq. km making it as Greater Chennai Corporation.

The entire area is classified into three administrative divisions namely North, South and Central Chennai with a total of 16 taluks. According to 2011 census, the population was about 6.7 million and ranks sixth in the country.

Two major rivers flow through Chennai, the Koovam through the centre and the Adyar River to the south. A third river, the Kosasthalaiyar, travels through the northern fringes of the city before draining into the Bay of Bengal, at Ennore. The Coastline extends up-to 58 km with Marina Beach as its longest.

3.4 ADMINISTRATIVE DIVISIONS OF CHENNAI

The entire area is classified into three administrative divisions namely North, South and Central Chennai with a total of 16 taluks is mentioned in the TABLE 3.1.

TABLE 3.1 ADMINISTRATIVE DIVISIONS OF CHENNAI

S.NO	TALUK	AREA (sq.km)
1	Alandur	26 km ²
2	Ambattur	36.56 km ²
3	Aminjikarai	16.49 km ²
4	Ayanavaram	15.3173 km ²
5	Egmore	17.4787 km ²
6	Guindy	20.1023 km ²
7	Madhavaram	34.62 km ²
8	Maduravoyal	22.65 km ²
9	Mambalam	14.1581 km ²
10	Mylapore	22.3171 km ²
11	Perambur	16.1593 km ²
12	Purasaiwakam	16.797 km ²
13	Shollinganallur	75.5879 km ²
14	Thiruvottiyur	62.344 km ²
15	Tondiarpet	12.3793 km ²
16	Velacheri	18.599 km ²
	TOTAL	426.493 km²

3.5 TOPOGRAPHY OF THE STUDY AREA

Being located on the Eastern coastal plain, most part of the city has a flat topography with a very gentle slope towards the East. The city's average elevation is around 6.7 metres. The highest elevation is noted in the areas of St.Thomas mount and Pallavaram.

3.6 CLIMATE OF THE STUDY AREA

The City has a mix of tropical wet and dry climate with 15.8°C as its lowest and 45°C as its highest. The months of March to June are typically the hottest. There will be less amount or even no precipitation during these months. They receive most of the rainfall from the North-East monsoon winds from October to December. On an average it receives a reasonable rainfall of 140 cm and at times it may exceed the limits.

3.7 CONCLUSION

The study area and its description have been explained in this chapter.

CHAPTER 4

MATERIAL AND METHODOLOGY

4.1 GENRAL

This chapter deals with the materials, methodology and software that have been used in this study.

4.2 DATA USED

The data used for assessing the cyclone risk are shown in the
TABLE 4.1

TABLE 4.1 DATA USED

S.NO	DATA	SOURCE
1	Cyclone windspeed	NASA Data Access Viewer
2	Cyclone Frequency	International Best Track Archive for Climate Stewardship (IBTrACS)
3	Rainfall Intensity	High Resolution Grided Datasets
4	Slope	USGS – Earth Explorer
5	Elevation	USGS – Earth Explorer
6	Land Use Land Cover	ESA Copernicus Open Access Viewer
7	Cyclone Track Range	International Best Track Archive for Climate Stewardship (IBTrACS)
8	Coastline Range	Open Street Map
9	Stream Range	BBBike Extracts Openstreet Map
10	Population Density	GeoIQ
11	Household Density	BBBike Extracts Openstreet Map
12	Vegetation Cover	Google Earth Pro

4.3 DATA DESCRIPTION

4.3.1 DEM

A Digital Elevation Model (DEM) is a 3-Dimensional representation of the topography of a land surface or terrain. DEMs are often used in GIS and are the most common basis for digitally produced relief maps. In the study, we used SRTM (Shuttle Radar Topography Mission) DEM with a 30m spatial resolution to perform slope and elevation. The data was downloaded from the USGS Earth Explorer. SRTM provides elevation values with high precision, allowing for reliable slope calculations.

4.3.2 Sentinel - 2

Sentinel-2 is an Earth Observation mission from the Copernicus program that systematically acquires optical imagery at high spatial resolution (10 m to 60 m) over land and coastal waters. The mission currently operates two satellite Sentinel 2A and Sentinel 2B. A third satellite, Sentinel-2C is undergoing testing in preparation for launch in 2024. The Sentinel 2 mission comprises a constellation of 2 polar orbiting satellites placed in the same sun-synchronous orbit, phased at 180° to each other.

In this study, we used Sentinel 2A data with a 10m resolution to identify the changes in land use land cover patterns. This data was downloaded from ESA Copernicus Open Access Viewer. In total it has 13 multispectral bands in which we used band 8, band 3 and band 2. Sentinel-2A captures data in multiple spectral bands, including visible, near-infrared, and shortwave infrared bands. Thus the information of the Spectral Bands of Sentinel 2A are shown in the TABLE 4.2

TABLE 4.2 SPECIFICATONS OF SENTINEL 2A

Spectral Band	Resolution (m)	Wavelength (nm)	Description
B1	60m	443nm	Ultra Blue (Coastal and Aerosol)
B2	10m	490nm	Blue
B3	10m	560nm	Green
B4	10m	665nm	Red
B5	20m	705nm	Visible and Near Infrared (VNIR)
B6	20m	740nm	Visible and Near Infrared (VNIR)
B7	20m	783nm	Visible and Near Infrared (VNIR)
B8	10m	842nm	Visible and Near Infrared (VNIR)
B8a	20m	865nm	Visible and Near Infrared (VNIR)
B9	60m	940nm	Short Wave Infrared (SWIR)
B10	60m	1375nm	Short Wave Infrared (SWIR)
B11	20m	1610nm	Short Wave Infrared (SWIR)
B12	20m	2190nm	Short Wave Infrared (SWIR)

4.3.3 CYCLONE TRACK

The Cyclone track details and frequency of occurrences are collected from International Best Track Archive for Climate Stewardship (IBTrACS) provided by National Oceanic and Atmospheric Administration (NOAA). The International Best Track Archive for Climate Stewardship (IBTrACS) project is the most complete global collection of tropical cyclones available. It provides data in a centralized location to aid our understanding of the distribution, frequency, and intensity of tropical cyclones worldwide. The data spans from 1840s to present, generally providing data at 3-hour intervals.

4.3.4 COASTLINE

The coastline data was obtained from Open Street Map (OSM) for our study area, to find the distance from the coastline to land. OSM is free, open, and global community project. Its applications include routing and navigation, converting GPS coordinates to addresses, searching places by name, analyzing map data and many more. It is developed and added to by community mappers around the world.

4.3.5 STREAM

The stream data was obtained from BBBike extracts OSM for the study area, to find the distance from the stream. It provides global map data in a unified tagging schema, although there are some local variation. It allows you to extracts areas from Planet.osm in OSM, PBF, o5m, Garmin, Organic Maps, mbtiles, OsmAnd, Esri shapefile, mapsforge, GeoJSON, SQLite, text or CSV format. It is a route planner for cyclists in Berlin. It is now ported to other cities around the world.

4.3.6 CYCLONE WIND SPEED

The wind speed data was acquired from POWER Data Access Viewer provided by NASA. The POWER Data Access Viewer (DAV) Web Mapping Application contains geospatially enabled solar, meteorological, and cloud related parameters formulated for assessing and designing renewable energy systems.

4.3.7 RAINFALL

The rainfall data was downloaded from the website POWER Data access viewer provided by NASA. The time period is from the year 2011 - 2020. The POWER Data Access Viewer (DAV) Web Mapping Application contains geospatially enabled solar, meteorological, and cloud related parameters formulated for assessing and designing renewable energy systems.

4.3.8 POPULATION DATA

The population data was acquired from GeoIQ. It is a platform developed by ESRI, a leading provider of Geographic Information System (GIS) software and solutions. It was designed to enable users to easily create, share, and analyze geospatial data and maps online.

4.3.9 HOUSEHOLD DATA

The Household data was obtained from BBBike extracts OSM for the study area, to find the distance from the stream. It provides global map data in a unified tagging schema, although there are some local variations. It allows you to extract areas from Planet.osm in OSM, PBF, o5m, Garmin, Organic Maps, mbtiles, OsmAnd, Esri shapefile, mapsforge, GeoJSON, SQLite, text or CSV format. It is a route planner for cyclists in Berlin. It is

now ported to other cities around the world.

4.3.10 VEGETATION COVER

The Vegetation cover data was acquired from Google Earth Pro which is in polygon format. Google Earth Pro was a professional version of Google Earth, a virtual globe, map, and geographical information program that was initially released by Google in 2001. The combination of high-resolution imagery, measurement tools, historical data access, GIS data import capabilities, customization options, export capabilities, and user-friendly interface made Google Earth Pro makes it a valuable tool for extracting vegetation cover information and conducting vegetation-related analyses.

4.4 SOFTWARE USED

GIS software are computer-based programs and tools that allows the user to visualize, analyze, interpret and store the geographic datasets. The software that has been for the study are,

1. ArcGIS Desktop
2. ENVI 5.3
3. Google Earth Pro
4. MS Excel

4.4.1 ARCGIS DESKTOP

ArcGIS is a Desktop software developed by ESRI which can be used primarily to view, edit, create, and analyze geospatial datasets. ArcMAP 10.8 has been used in particular, which is a part of the ArcGIS desktop suite.

ArcMAP is the former main component of ESRI's ArcGIS suite of geospatial processing programs. ArcMap allows the user to explore data within a data set, symbolize features accordingly, and create maps. The ArcMap interface has two main sections, including the table of contents on the left and the data frames which display the map. Items in the table of contents correspond with the layers on the map.

4.4.2 ENVI 5.3

ENVI (Environment for Visualizing Imagery). ENVI 5.3 is an advanced software tool designed for Remote Sensing analysis and image processing. It is developed by Harris Geospatial Solutions and is widely used by researchers, scientists, geospatial professionals, and government agencies for a variety of applications including environmental monitoring, land cover classification, agriculture, forestry, urban planning, and disaster management.

ENVI 5.3 allows users to analyse various types of satellite and aerial imagery, including multispectral, hyperspectral, LiDAR, and radar data. It provides wide range of tools and algorithms for image enhancement, visualization, classification, feature extraction, and change detection.

4.4.3 GOOGLE EARTH PRO

Google Earth was originally developed at Intrinsic Graphics in the late 1990s. Google Earth Pro is a powerful geospatial mapping tool developed by Google. Designed to provide the requirements of professionals, researchers, and enthusiasts. Google Earth Pro offers a suite of advanced features for complex spatial analysis and visualization tasks. Google Earth provides a series of other tools through the desktop application, including a measure distance tool.

4.4.4 MS-Excel

Microsoft Excel has the basic features of all spreadsheets, using a grid of cells arranged in numbered rows and letter-named columns to organize data manipulations like arithmetic operations. It has a battery of supplied functions to answer statistical, engineering and financial needs. In addition, it can display data as line graphs, histograms and charts with a very limited three-dimensional graphical display. Spreadsheet applications such as MS Excel use a collection of cells arranged into rows and columns to organize and manipulate data. It permits users to arrange data in order to view various factors from different perspectives.

4.5 METHODOLOGY

The methodology mainly focusses in identification of Risk zones in our study area.

The flowchart of methodology is shown in FIGURE 4.1

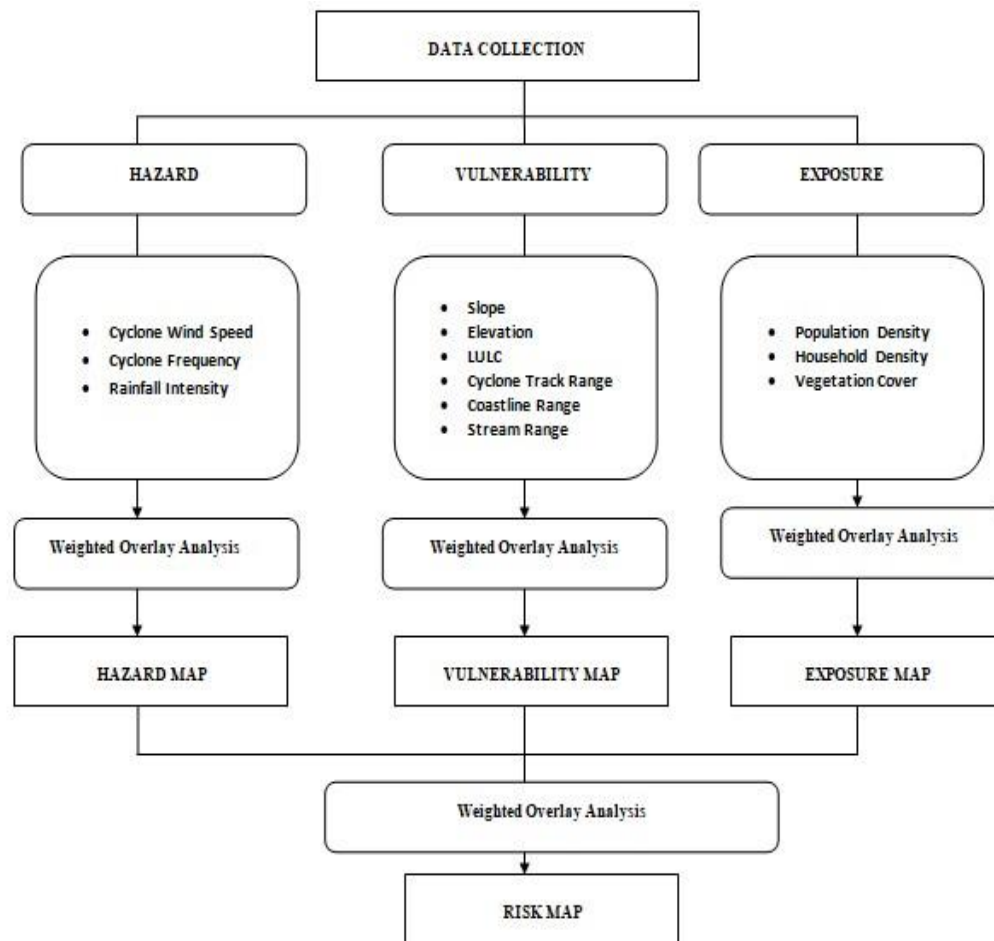


FIGURE 4.1 FLOW CHART OF THE METHODOLOGY

4.6 CRITERIA FOR RISK ASSESSMENT

Risk results from the combination of hazards, conditions of vulnerability and level of Exposure. In a total of 13 criterion are associated with hazard, vulnerability and exposure. The appropriate selection of different criteria under each parameter makes the risk assessment more accurate.

$$\text{Risk} = \text{Hazard} \times \text{Exposure} \times \text{Vulnerability}$$

4.6.1 CRITERIA FOR HAZARD

A Hazard is a potentially damaging physical event or phenomenon that may cause loss of life or injury, property damage, social and economic disruption or environmental degradation. This event has a probability of occurrence within a specified period of time and within a given area, and has a given intensity. Since the study focuses on tropical Cyclone, the criteria for Hazard includes Cyclone wind speed, Cyclone frequency and Rainfall intensity.

CYCLONE WIND SPEED

Cyclone wind speed can be defined as the rate at which wind travels through a specific area during a cyclone. It is one of the most important factors in determining the potential damage of cyclone. Cyclone wind speed determines the intensity of cyclone, it means how much destruction or damage can happen on the human habitation and natural resources.

To show the spatial distribution of wind speed, a spatial layer of cyclone windspeed was created based on previous cyclone event i.e., Severe Cyclonic Storm Michaung (1-6 December, 2023). The data on windspeed were collected from the NASA Power Data Access Viewer. IDW interpolation technique was applied using ArcGIS to prepare Cyclone wind speed map.

CYCLONE FREQUENCY

Cyclone frequency refers to number of cyclones occur in a particular area or region. It's a measure of how many cyclones happen in a certain amount of time, like a year or a decade. The frequency of tropical cyclones is measured by the number of cyclones occurring in a given area over a specific period. If a region has a high cyclone frequency, it means that cyclones happen there often, whereas a low cyclone frequency means they happen less often.

In this study, Historical Cyclone Records were acquired from NOAA's National Centers for Environmental Information/ International Best Track Archive or Climate Stewardship (IBTrACS). To prepare a spatial layer of cyclone frequency, the earlier cyclone records from the period of 1890–2023 and their effects on the study region were taken into consideration. Earlier records indicate that different parts of the study region had suffered by 13 cyclonic storms over all the years. The spatial analyst tool of ArcGIS (version 10.8) was used to convert the cyclone frequency data into a spatial layer.

RAINFALL INTENSITY

Rainfall intensity is defined as the ratio of the total amount of rain falling during a given period at a specific location. It is expressed in depth units per unit time, usually as mm per hour (mm/h). The precipitation pattern of the study area strongly influences to intensify the impacts of a tropical cyclone. Tropical cyclones (TCs) are associated with heavy rainfall. Tropical cyclones cause extreme rainfall that ultimately intensifies flooding in the affected area.

In this study, yearly rainfall data were acquired from the NASA Power data access viewer from 2011 to 2020 to prepare the rainfall intensity map. Here IDW interpolation technique was applied using ArcGIS to create rainfall intensity map.

4.6.2 CRITERIA FOR VULNERABILITY

Vulnerability is the inability to resist or respond when a disaster occurs. It can be of physical, social, economical and environmental vulnerability. It varies from individuals, groups, communities and can be different for different locations. The six selected criterion includes slope, elevation, LULC, cyclone track range, coastline range, stream range.

SLOPE

Slope refers to the changes in the physical features of the land with its elevation, orientation and topography. Slope can be expressed either in degree or percentage. This study utilized the SRTM DEM data with 30m spatial resolution, which was obtained from USGS Earth Explorer. Using spatial analyst tool in ArcGIS, the slope for the study area is created.

Areas with a steep slope are generally less vulnerable to cyclones. In contrast, areas with low elevation with a gentle slope are highly vulnerable to cyclones. Therefore, slope of any location is to show the maximum rate of variation in between that location and its surrounding area.

ELEVATION

Elevation refers to the height of an object on the Earth's surface from the Mean Sea Level (MSL). Elevations are usually measured in meters or feet. This study utilized the SRTM DEM data with 30m resolution, which was obtained from USGS Earth Explorer. As DEM data is a Binary data it is converted into Raster data for the next process. By adjusting the symbology to display elevation values as a continuous surface, we get elevation for the study area.

Areas with high elevation are less vulnerable to cyclone since they act as a barrier whereas areas with low elevation are highly vulnerable to cyclone. Elevation is considered as an important parameter of Vulnerability.

LULC

Land cover is simply what covers the earth's surface and land use describes how the land is used. Land Use and Land Cover (LULC) is a valuable exp indicator for tropical cyclone vulnerability assessment.

In this study, Sentinel-2 data with a 10 m spatial resolution was used to generate the LULC map. In ArcGIS, the following bands 8, 4 and 3 of the sentinel 2A data were used for the False Color Composition (FCC). This FCC is then imported in ENVI 5.3, using region of interest from basic tools, the training samples were classified according to the land use and land cover for the study area. The ROI were classified into Settlements, Waterbody, Vegetation, Barren land and Wet Land. Then the Supervised classification and Maximum likelihood classification were performed to get the accuracy for the land use and land cover. Among these Settlement are considered as important as they are highly affected during Cyclones.

CYCLONE TRACK RANGE

A Cyclone track refers to the path of a cyclone's center over a specific period, typically displayed on a map or chart. It shows the cyclone's movement, direction, and speed, helping forecasters and researchers understand its behavior and potential impact. Areas within a 100-km radius of the cyclone's center face a higher risk of damage. The proximity of a region to the path of a cyclonic system plays a critical role in determining its vulnerability to cyclones.

The study used the data of NOAA's (National Centers for Environmental Information) International Best Track Archive for Climate Stewardship (IBTrACS) and a buffer was created for mapping the distance from the cyclone track.

COASTLINE RANGE

The boundary of a coast, where land meets water, is called the coastline. Waves, tides, and currents help to create coastlines. Coastline is considered as important factor for assessing cyclone vulnerability.

In this study, Coastline data was obtained from Open Street Map. The Coastline range was assessed by calculating the distance from the coastline utilizing the ruler tool in Google Earth Pro and subsequently, the proximity to coastline was generated. The areas near the coast are high vulnerability to cyclones than the areas that are far away from the coast.

STREAM RANGE

A stream is a continuous body of surface water flowing within the bed and banks of a channel. Cyclone may have a significant negative influence on human life as well as property that are situated close to a major water body.

For the study, the stream network data is downloaded from the Open Street Map. A Multiring buffer was created for the stream and the regions near the stream are highly vulnerable due to flooding events.

4.6.3 CRITERIA FOR EXPOSURE

Exposure indicates the degree to which the elements-at-risk are exposed to a particular disaster. It may be people, property, infrastructure etc. The different demographics such as age groups, sex, number of people, profession and type of assets would have different levels of vulnerability. In this study the criteria for exposure includes population density, household density and vegetation cover.

POPULATION DENSITY

The number of individuals living within that specific area determines the population density, or the number of individuals divided by the size of the area. Population density is used as an indicator of exposure, which is one of the components of risk assessment, along with hazard and vulnerability.

In this study, the population density layer was prepared on the basis of census data. Areas with high population density, such as major cities and coastal regions, are considered to be at high risk due to the large number of people and assets exposed to cyclone hazards

HOUSEHOLD DENSITY

Household density refers to the number of houses per unit area, typically measured in households per square kilometer or square mile. It is used to estimate the number of people and households potentially affected by cyclone hazards, such as storm surges, heavy rainfall, and strong winds.

In this study, Household data were extracted from BBBike extracts Open Street Map Website. We prepared the Household Density Layer by using the household data collected. Areas with high household density are more vulnerable to cyclone hazards due to the concentration of people, buildings, and infrastructure.

VEGETATION COVER

Vegetation cover plays a crucial role in exposure assessment. Vegetation cover is a key factor in land cover classification. Vegetation cover is highly exposed to Cyclone after the Settlement areas.

In this study, vegetation cover layer was prepared by identifying the vegetation using Google Earth Pro, and further by performing the Interpolation Method.

4.7 WEIGHTED OVERLAY

The weighted overlay is a commonly used technique in remote sensing for combining multiple layers of data into a single composite layer.

It is a method of using a common scale to different inputs to find the integrated result. This technique assigns weights to each input layer based on its relative importance and combines them using a weighted sum approach.

This technique is applied separately to each spatial criteria layer of Hazard, Vulnerability and Exposure.

The steps involved in weighted overlay are

- 1) Selection of input layers
- 2) Weight assignment
- 3) Rescaling of input layers
- 4) Weighted overlay.

4.8 CONCLUSION

The materials and methodology of the study have been explained in this chapter.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 GENERAL

This chapter deals with the discussion on cyclone risk mapping for greater Chennai, Tamil Nadu.

5.2 STUDY AREA EXTRACTION

The study area was extracted from Survey of India in shape file format is shown in FIGURE 5.1.

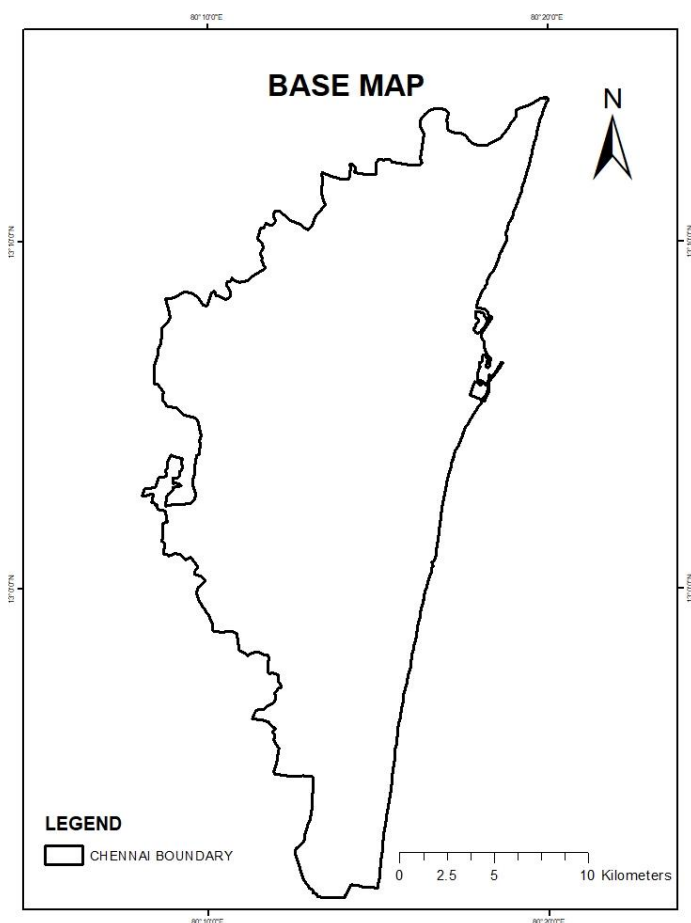


FIGURE 5.1 MAP OF STUDY AREA

5.3 CYCLONE WIND SPEED

Cyclone wind speed is the rate at which wind travels through a specific area. It is mentioned in km/h. The cyclone windspeed map is shown in FIGURE 5.2.

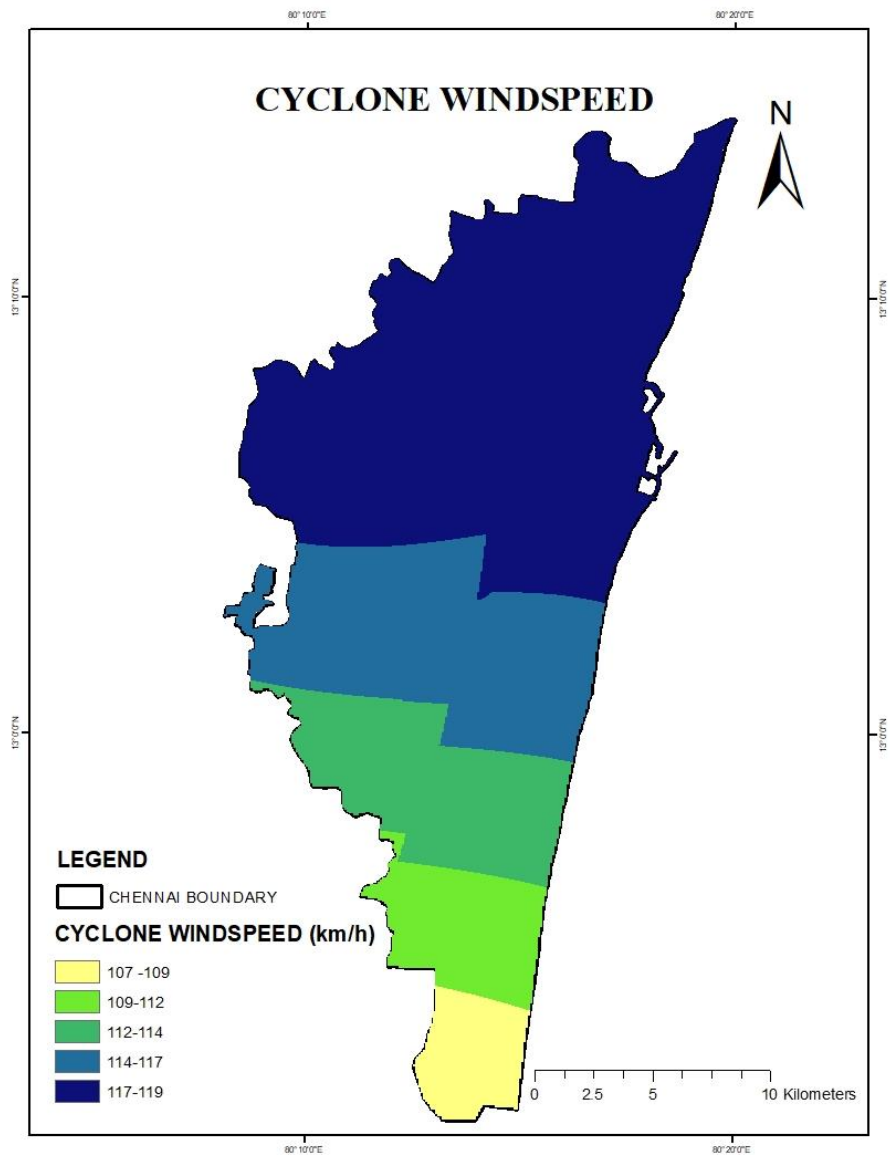


FIGURE 5.2 CYCLONE WINDSPEED MAP

5.4 CYCLONE FREQUENCY

Cyclone frequency refers to number of cyclones that occur in a particular area or region. It is a measure of how many cyclones happen in a certain amount of time, like a year or a decade. The cyclone frequency is shown in the FIGURE 5.3.

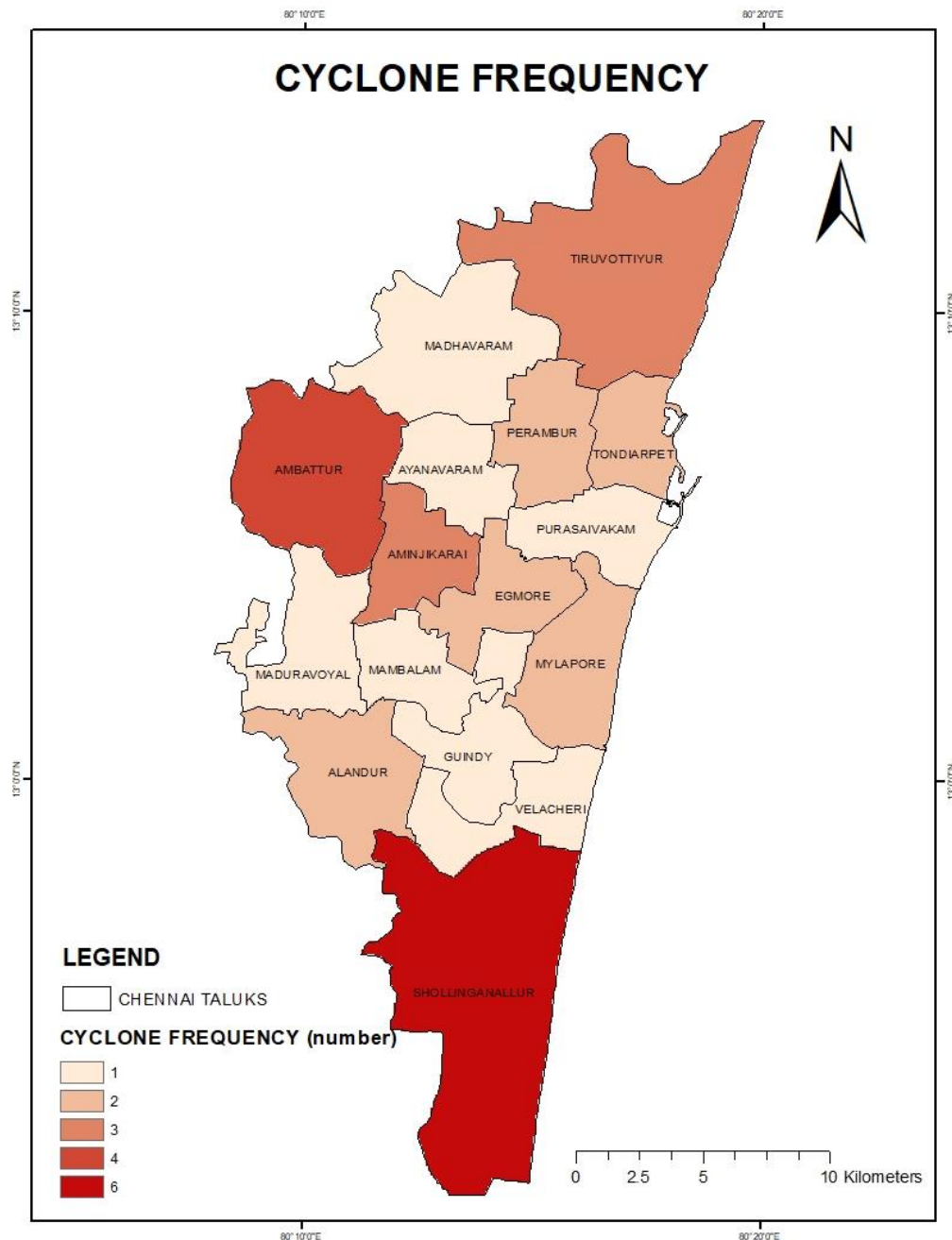


FIGURE 5.3 CYCLONE FREQUENCY MAP

5.5 RAINFALL INTENSITY

Rainfall intensity is defined as the ratio of the total amount of rain falling during a given period at a specific location. It is expressed usually in cm. The rainfall intensity map is shown in the FIGURE 5.4.

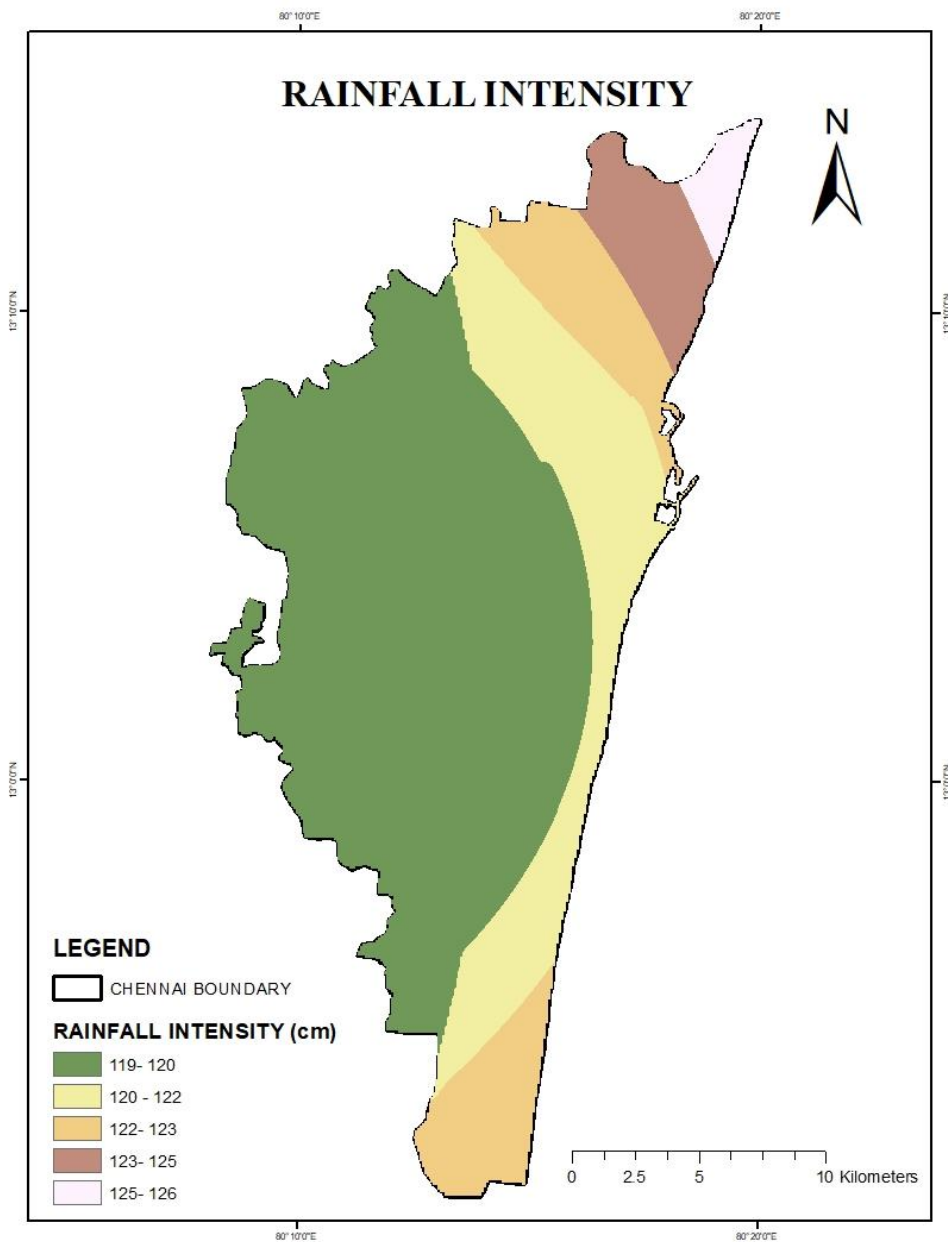


FIGURE 5.4 RAINFALL INTENSITY MAP

5.6 SLOPE

The Slope map of the study area was created using SRTM DEM. Areas with gentle slope are highly vulnerable to cyclones where areas with steep slope are less vulnerable. The slope map is shown in FIGURE 5.5.

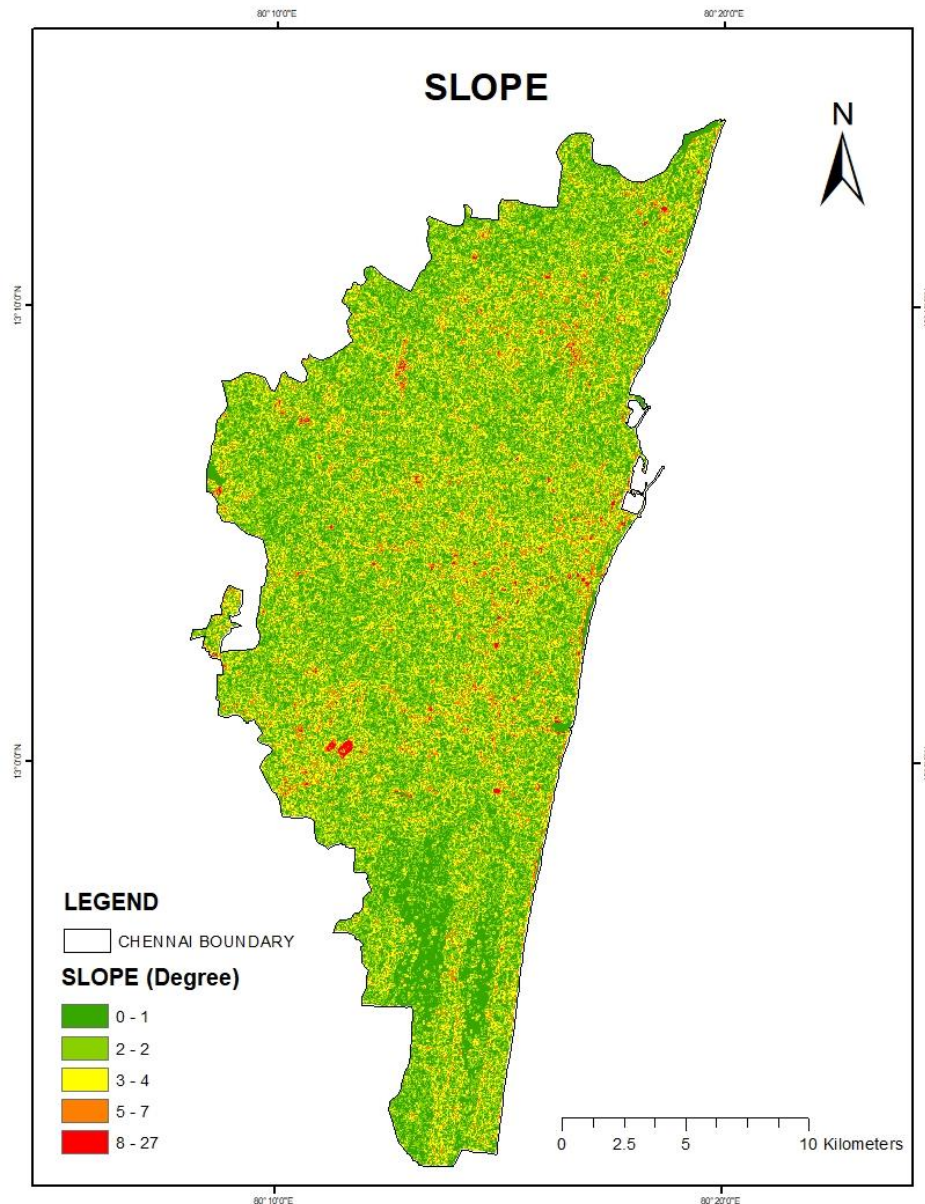


FIGURE 5.5 SLOPE MAP

5.7 ELEVATION

The Elevation refers to the height of the object on the Earth surface from the Mean Sea Level (MSL). The elevation map of the study area was created using SRTM DEM is shown in FIGURE 5.6.

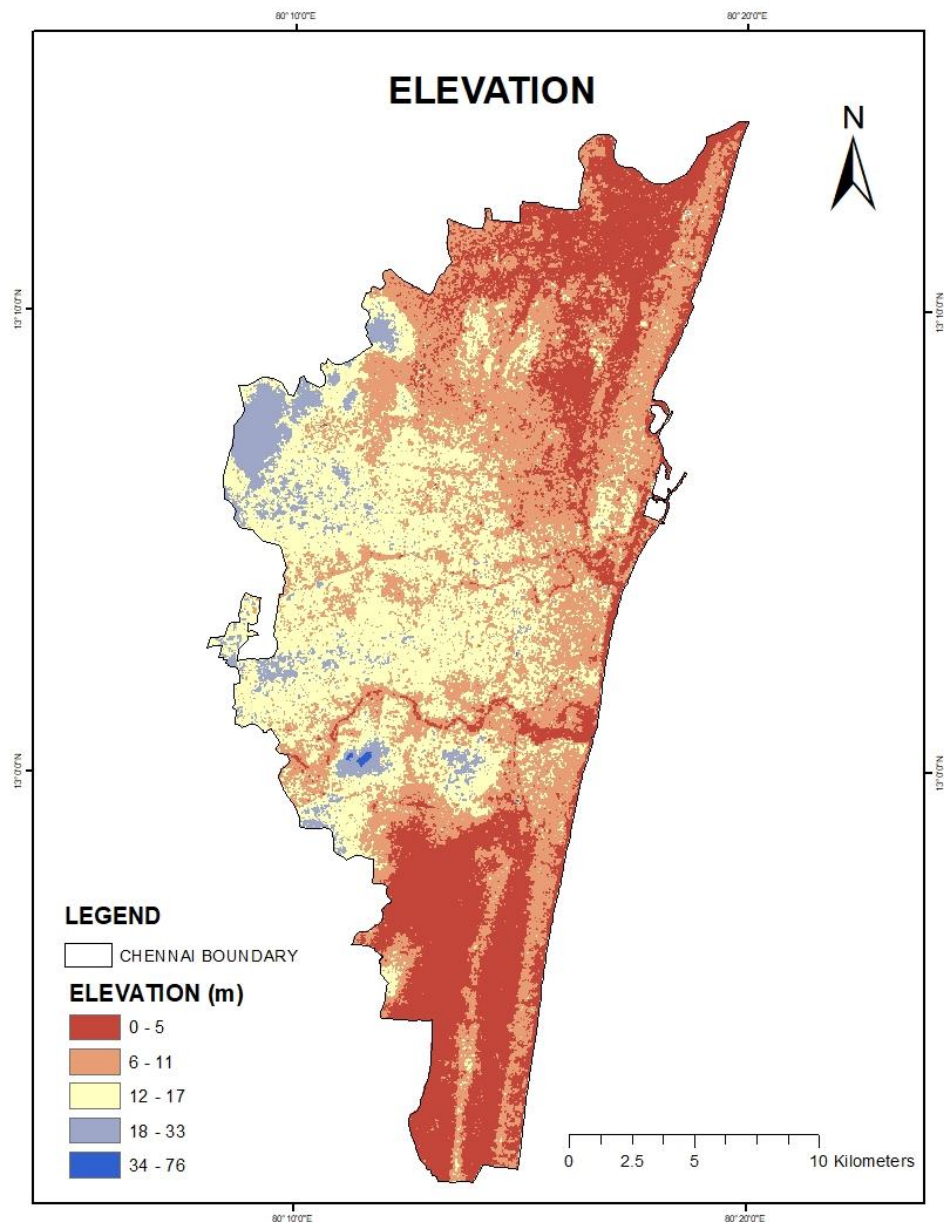


FIGURE 5.6 ELEVATION MAP

5.8 LAND USE LAND COVER

The Land Use and Land Cover of the study area is classified into 5 classes, namely settlements, vegetation, waterbody, barren land and wet land. The Land Use and Land Cover map for the study area is shown in the FIGURE 5.7

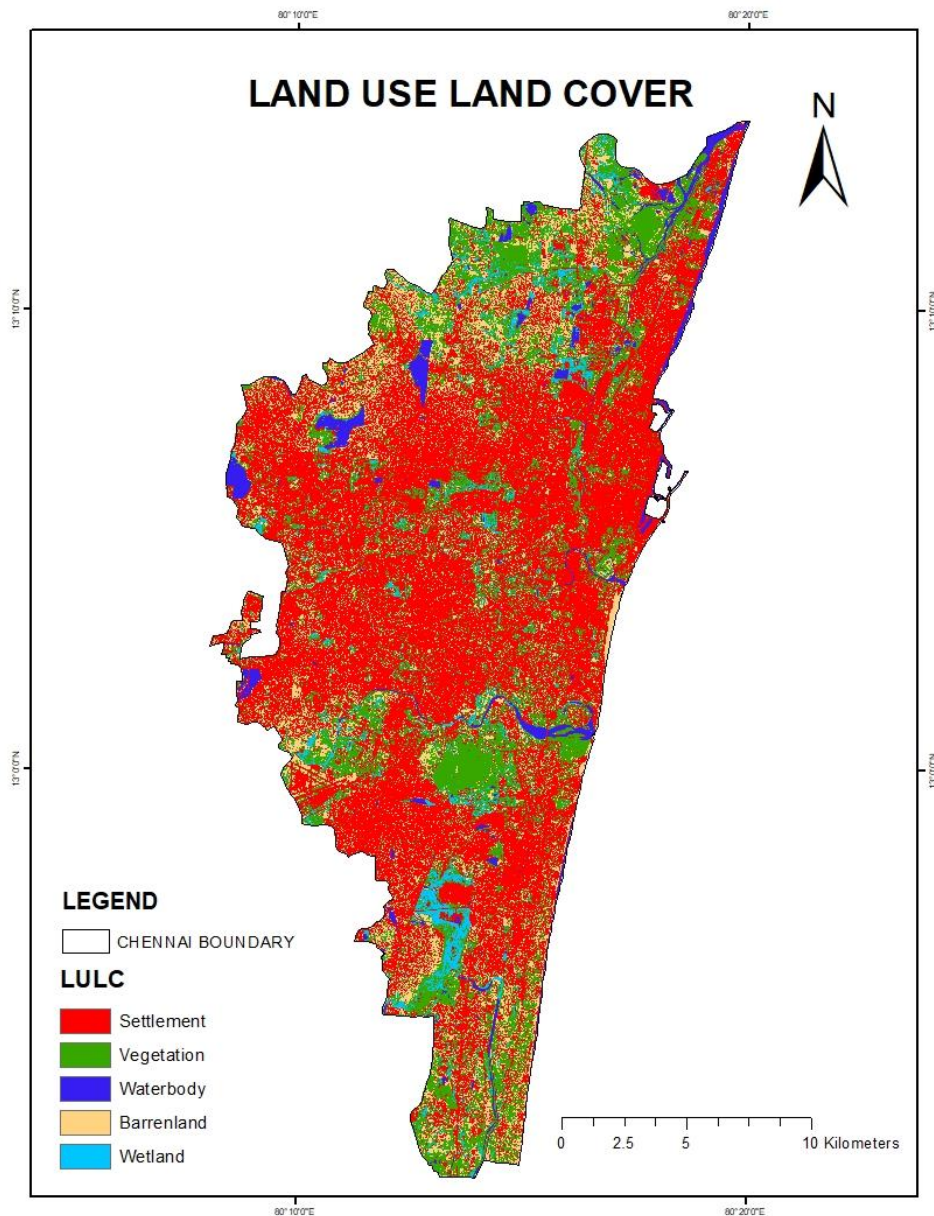


FIGURE 5.7 LAND USE LAND COVER MAP

5.9 CYCLONE TRACK RANGE

The Cyclone track range map of the study area is shown in the FIGURE 5.8

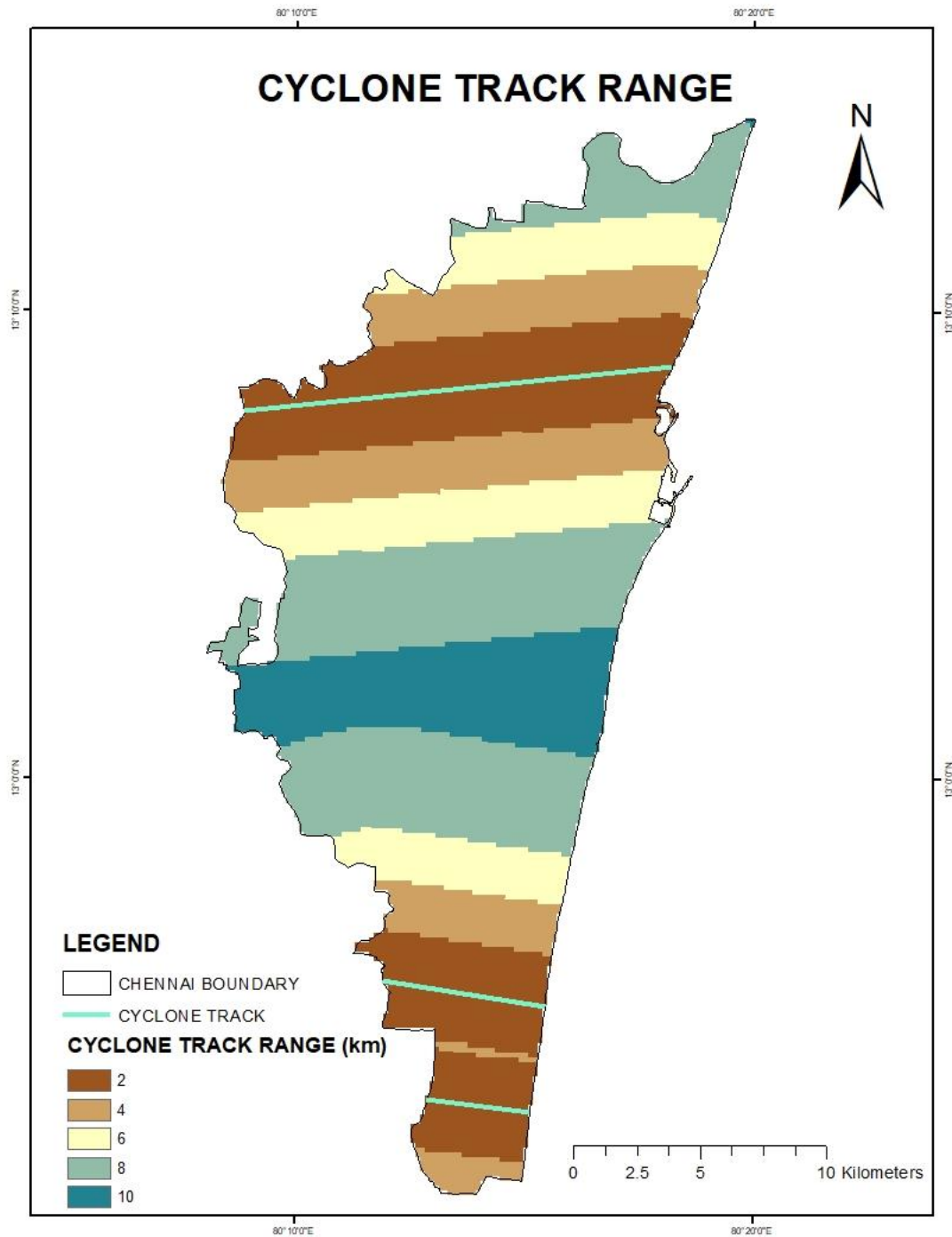


FIGURE 5.8 CYCLONE TRACK RANGE MAP

5.10 COASTLINE RANGE

The Coastline range map of the study area is shown in the FIGURE 5.9

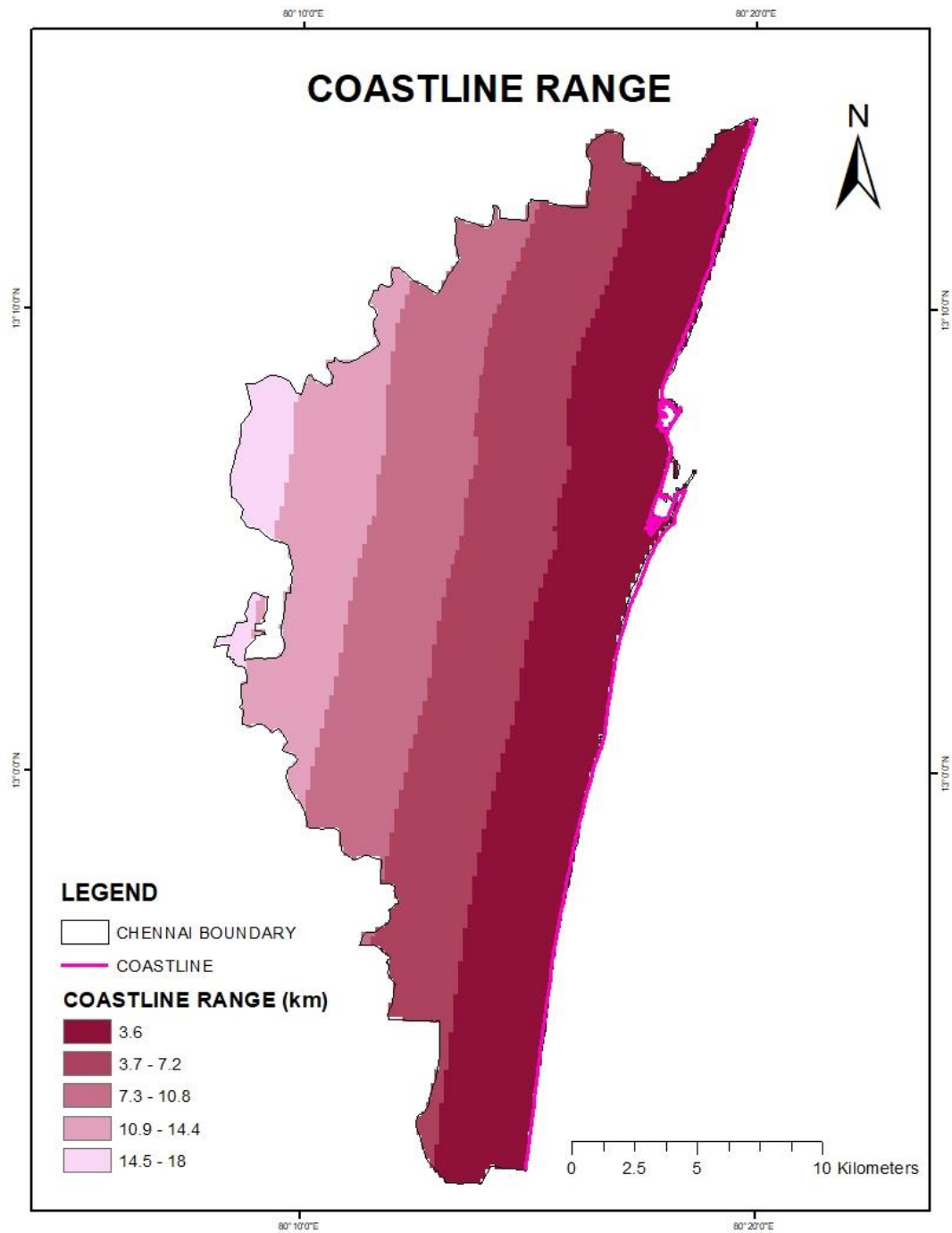


FIGURE 5.9 COASTLINE RANGE MAP

5.11 STREAM RANGE

The stream range map of the study area is shown in the FIGURE 5.10

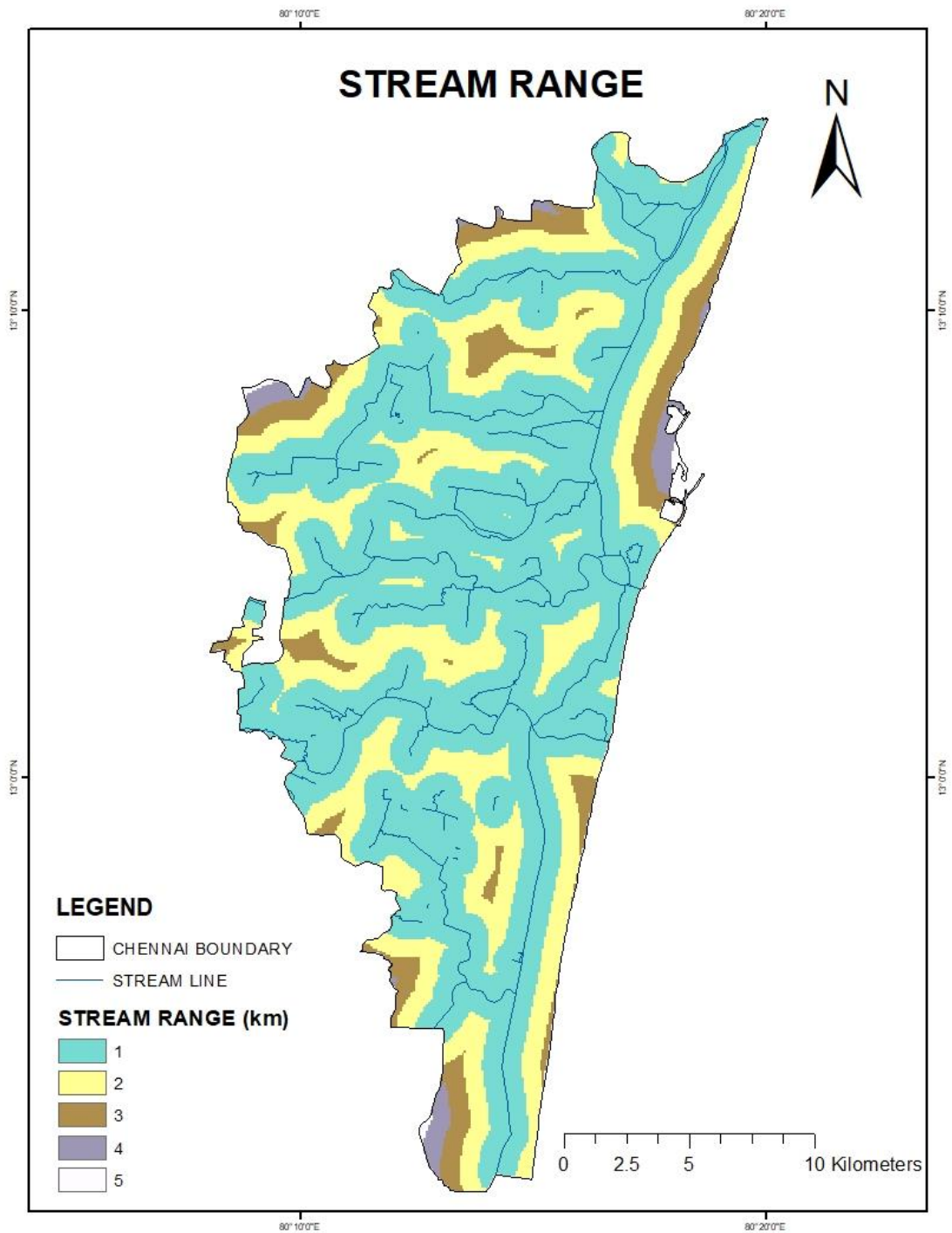


FIGURE 5.10 STREAM RANGE MAP

5.12 POPULATION DENSITY

The population density map of the study area map is shown in the FIGURE 5.11

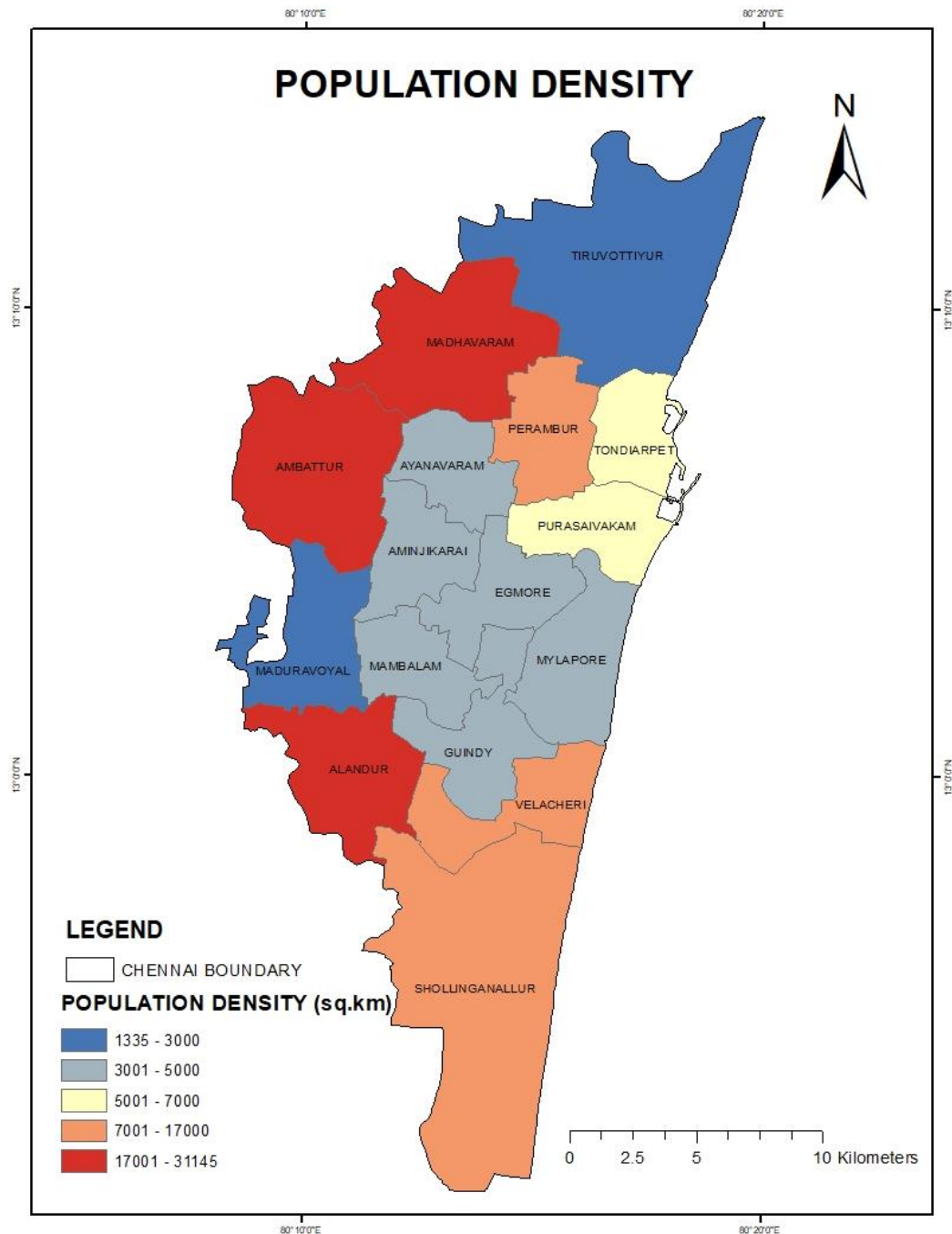


FIGURE 5.11 POPULATION DENSITY MAP

5.13 HOUSEHOLD DENSITY

The Household density map of the study area is shown in FIGURE 5.12

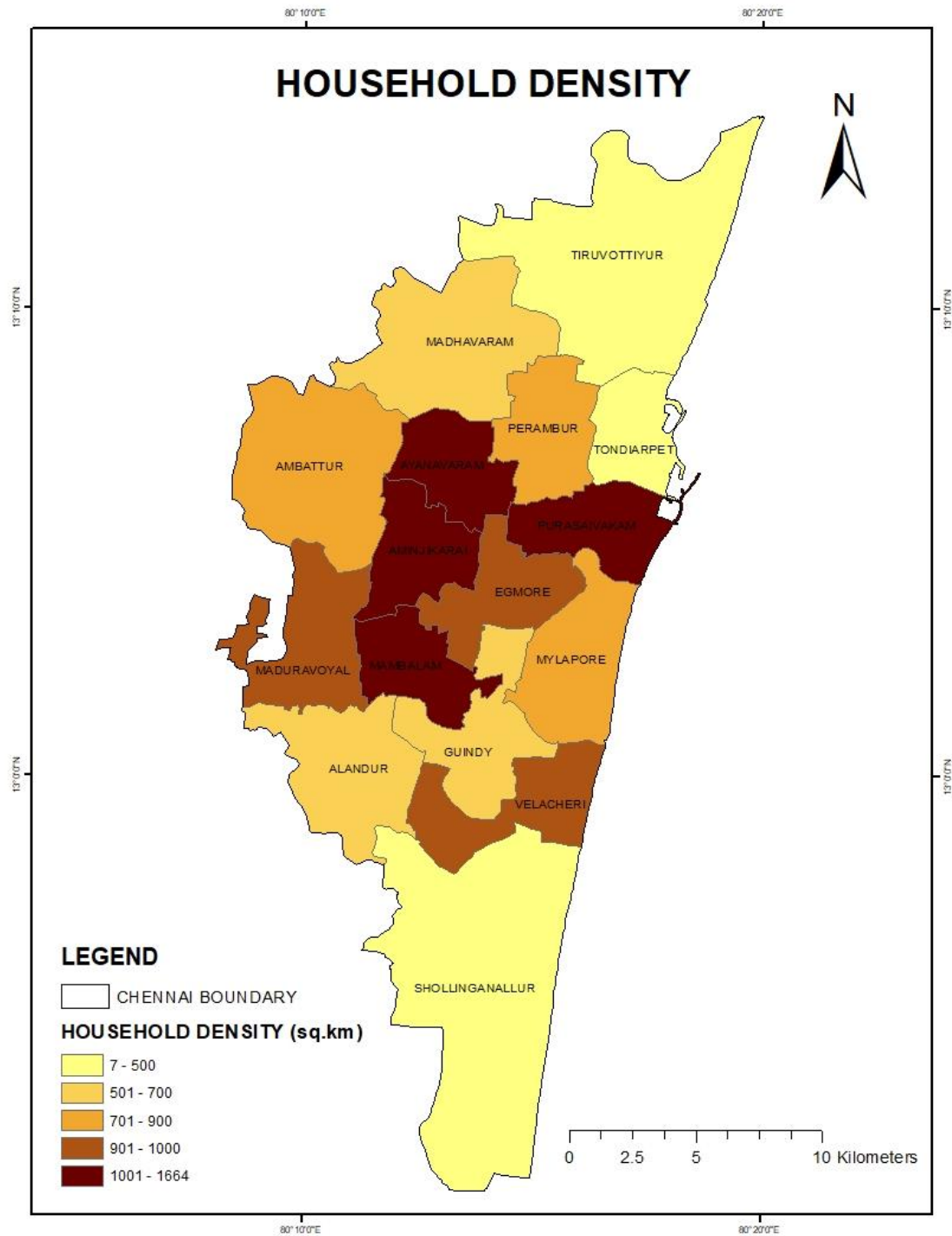


FIGURE 5.12 HOUSEHOLD DENSITY MAP

5.14 VEGETATION COVER

The Vegetation cover map of the study area is shown in the FIGURE 5.13

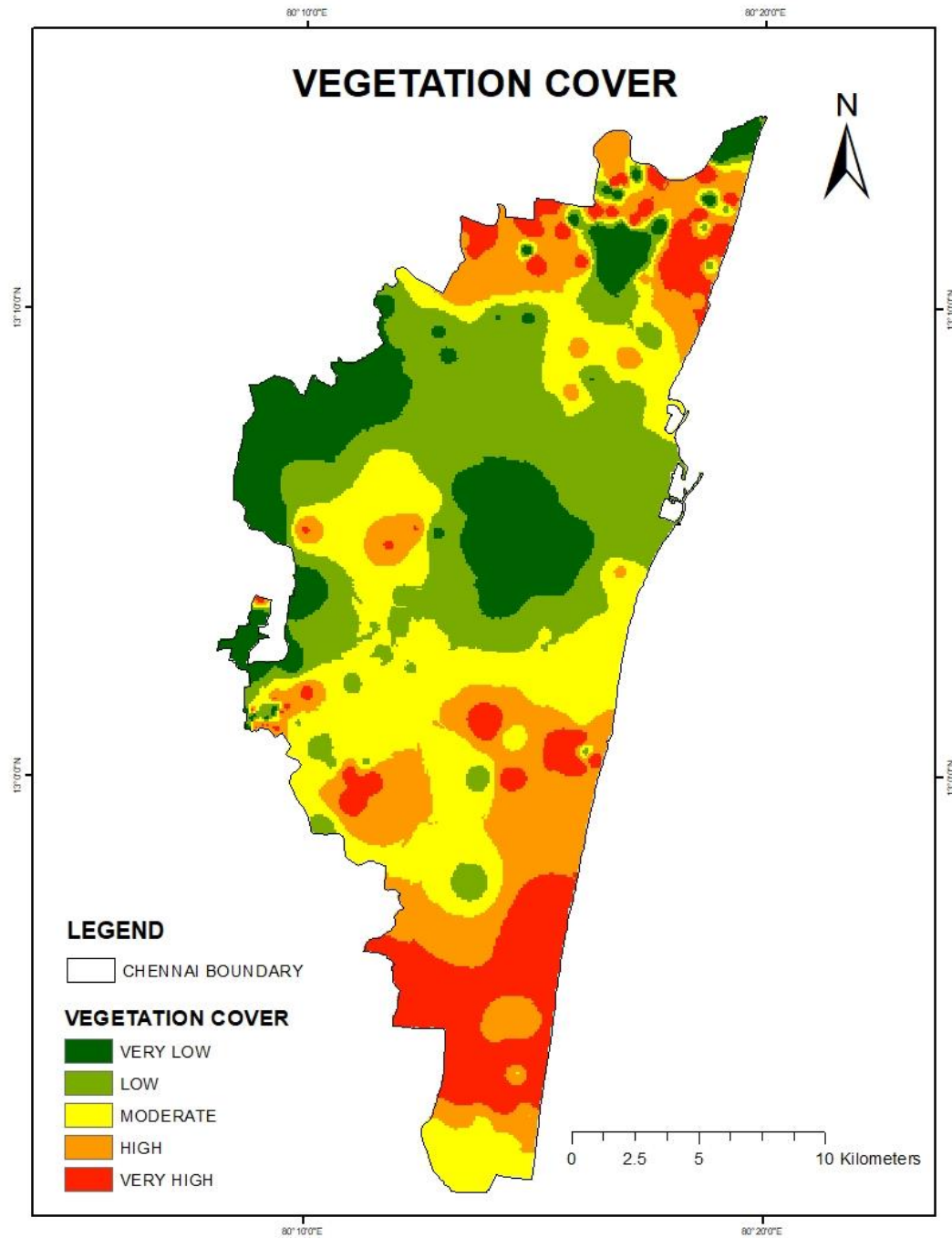


FIGURE 5.13 VEGETATION COVER MAP

5.15 HAZARD MAP

The Hazard map is developed by combining the individual maps of cyclone windspeed, cyclone frequency and rainfall intensity. The hazard map of the study area is shown in the FIGURE 5.14

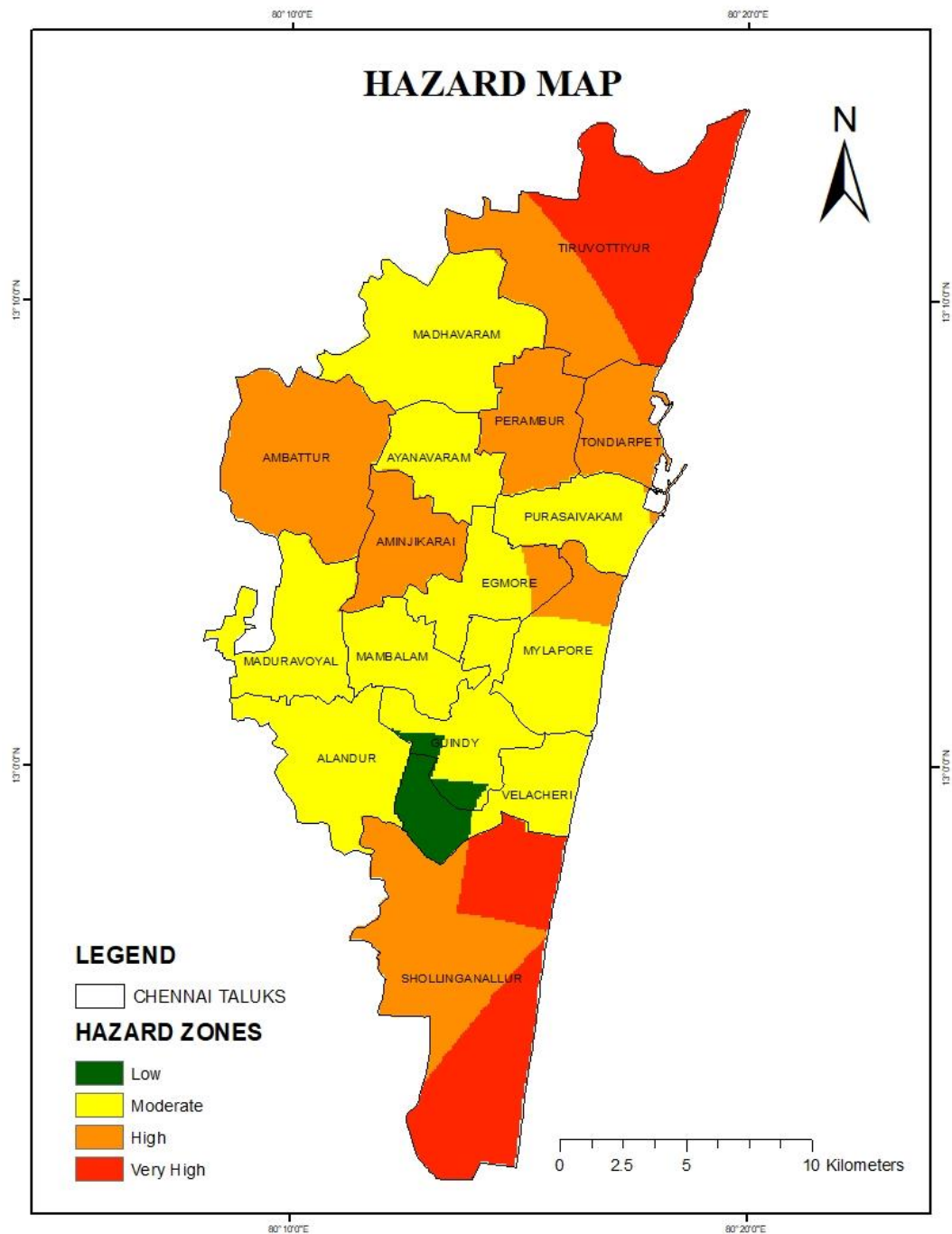


FIGURE 5.14 HAZARD MAP

**TABLE 5.1WEIGHTED OVERLAY ANALYSIS TABLE FOR HAZARD
MAP**

S.NO	PARAMETERS	VALUES	RANK	WEIGHTAGE (%)
1	CYCLONE WINDSPEED (km/h)	107 - 109 109 - 112 112 - 114 114 - 117 117 - 119	1 2 3 4 5	20
2	CYCLONE FREQUENCY (number)	1 2 3 4 6	1 2 3 4 5	35
3	RAINFALL INTENSITY (cm)	119 - 120 120 - 122 122 - 123 123 - 125 125 - 126	1 2 3 4 5	45

5.16 VULNERABILITY MAP

The Vulnerability map is developed by combining the individual maps of slope, elevation, LULC, cyclone track range, coastline range and stream range. The vulnerability map of the study area is shown in the FIGURE 5.15

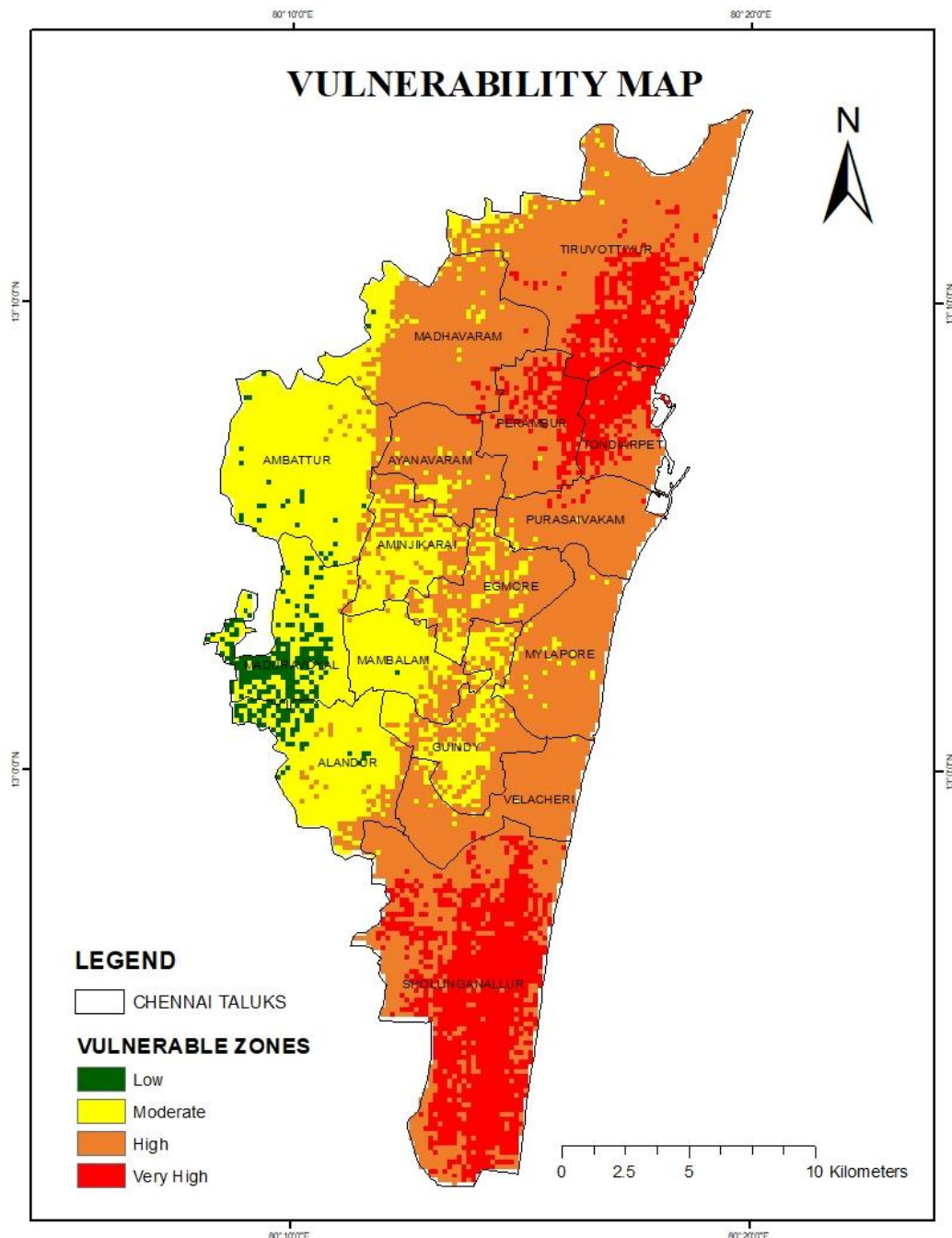


FIGURE 5.15 VULNERABILITY MAP

**TABLE 5.2 WEIGHTED OVERLAY ANALYSIS TABLE FOR
VULNERABILITY MAP**

S.NO	PARAMETERS	VALUES	RANK	WEIGHTAGE (%)
1	SLOPE (degree)	0 -1 2 - 2 3 - 4 5 - 7 8 - 27	5 4 3 2 1	10
2	ELEVATION (m)	0 - 5 6 - 11 12 - 17 18 - 33 34 - 76	5 4 3 2 1	20
3	LULC	SETTLEMENT VEGETATION WATERBODY BARRENLAND WETLAND	5 3 4 2 1	10
4	CYCLONE TRACK RANGE (km)	2 4 6 8 10	5 4 3 2 1	20
5	COASTLINE RANGE (km)	0 - 3.6 3.7 - 7.2 7.3 - 10.8 10.9 - 14.4 14.5 - 18	5 4 3 2 1	30
6	STREAM RANGE (km)	1 2 3 4 5	5 4 3 2 1	10

5.17 EXPOSURE MAP

The Exposure map is developed by combining the individual maps of population density, household density and vegetation cover. The Exposure map of the study area is shown in the FIGURE 5.16

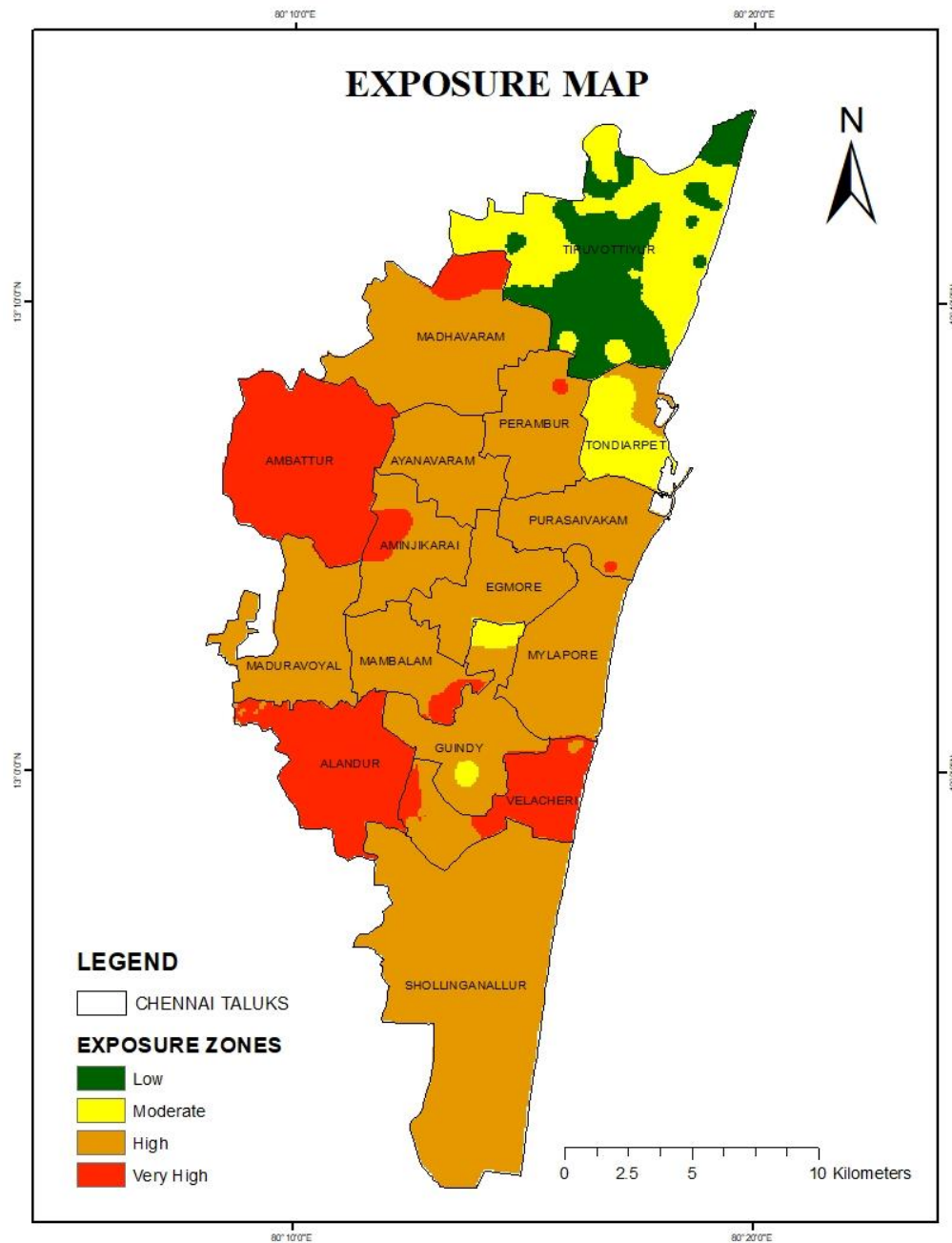


FIGURE 5.16 EXPOSURE MAP

**TABLE 5.3 WEIGHTED OVERLAY ANALYSIS TABLE FOR
EXPOSURE MAP**

S.NO	PARAMETERS	VALUES	RANK	WEIGHTAGE (%)
1	POPULATION DENSITY (sq.km)	1335 - 3000 3001 - 5000 5001 - 7000 7001 - 17000 17001 - 31145	1 2 3 4 5	40
2	HOUSEHOLD DENSITY (sq.km)	7 - 500 501 - 700 701 - 900 901 - 1000 1001 - 1664	1 2 3 4 5	40
3	VEGETATION COVER	VERY LOW LOW MODERATE HIGH VERY HIGH	1 2 3 4 5	20

5.18 RISK MAP

The Risk map is developed by integrating the maps of Hazard, Vulnerability and Exposure. The Risk map is shown in the FIGURE 5.17

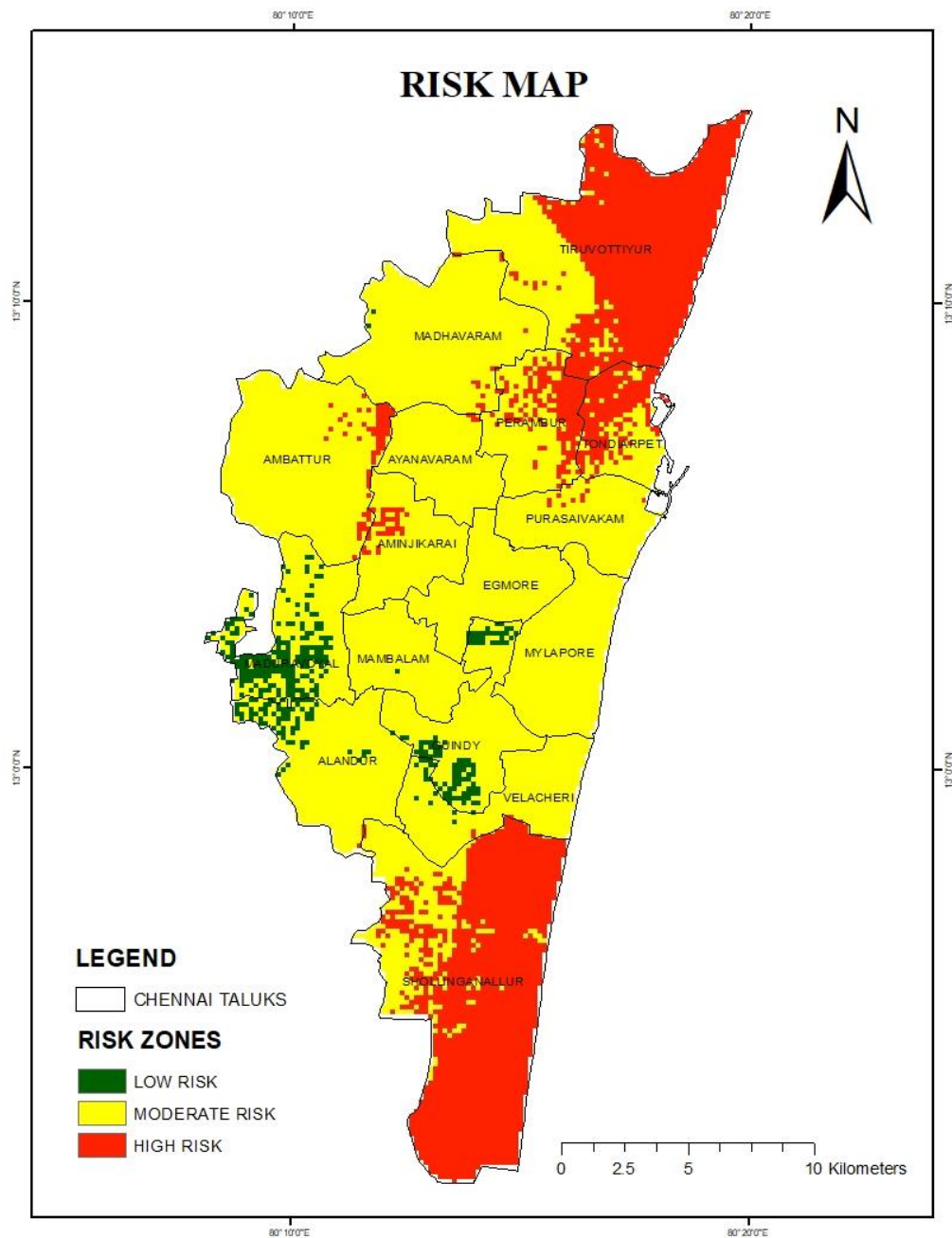


FIGURE 5.17 RISK MAP

TABLE 5.4 WEIGHTED OVERLAY ANALYSIS TABLE FOR RISK MAP

S.NO	PARAMETERS	VALUES	RANK	WEIGHTAGE (%)
1	HAZARD	LOW MODERATE HIGH VERY HIGH	1 2 3 4	40
2	VULNERABILITY	LOW MODERATE HIGH VERY HIGH	1 2 3 4	30
3	EXPOSURE	LOW MODERATE HIGH VERY HIGH	1 2 3 4	30

LINE GRAPH

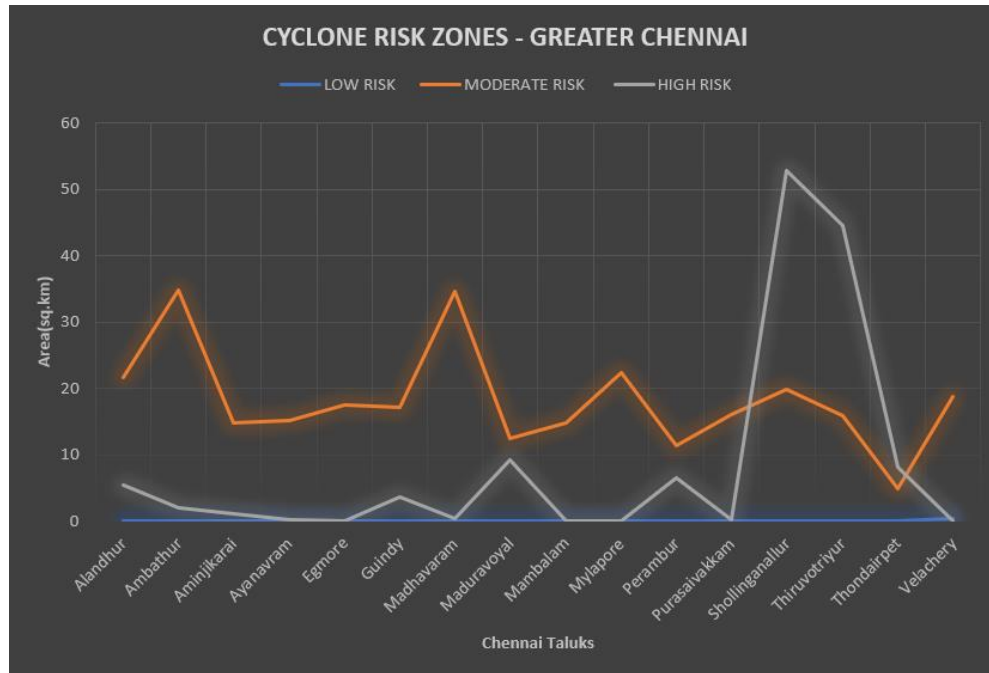


FIGURE 5.18 CYCLONE RISK ZONES LINE GRAPH

PIECHART FOR RISK MAP IN PERCENTAGE

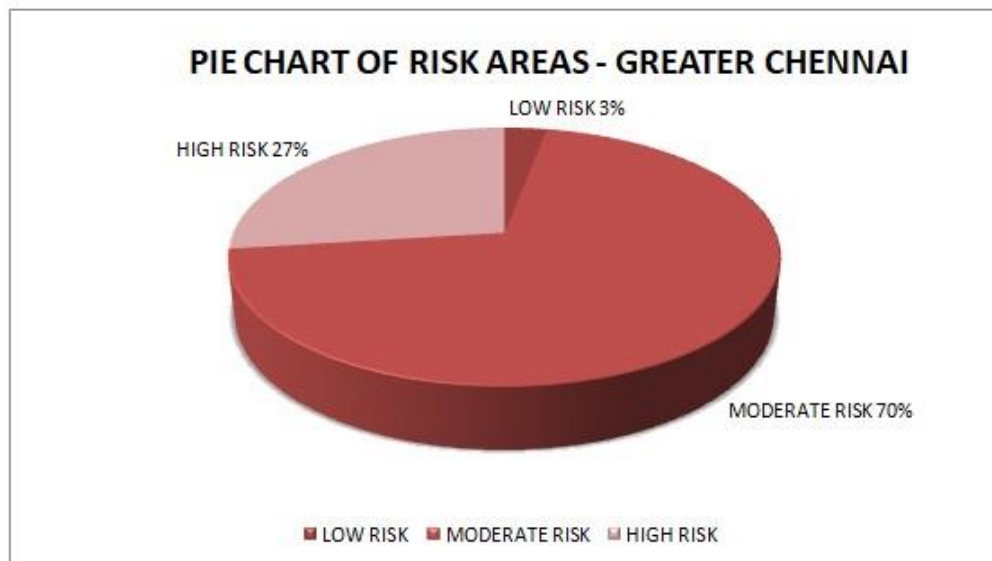


FIGURE 5.19 PIE CHART FOR THE RISK MAP IN PERCENTAGE

5.19 CONCLUSION

In this chapter, the results and discussions on the cyclone risk assessment for Greater Chennai have been discussed.

CHAPTER 6

CONCLUSIONS

6.1 CONCLUSION

The present study has assessed the Cyclone risk zones in Greater Chennai using Remote sensing and GIS techniques. The risk was assessed as the integration of hazard, vulnerability and exposure. A spatial layer was created for each risk component and then it was combined using Weighted Overlay method to produce the end risk map. The Weighted Overlay method is a commonly used method to combine multiple thematic layers to a single composite layer. The risk zones were divided into low, moderate and high.

The overall result shows that 27% of the total area comes under highly risk zone, 70% comes under moderate risk zone and 3% comes under lower risk zone. Hence the study revealed that more than half percent of the total area in Chennai comes under moderate risk zone. The result map will be useful for the local and coastal authorities in advance planning to reduce the potential negative consequences of tropical cyclone. The identification of risk zones in the study area provides an opportunity to increase the resilience of the risk communities to avoid loss of life and property.

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